

Probability Review

S : sample space

A : event (zarın 3'ten büyük olması)

a : outcome (zar = 2)

↳ every outcome is an event

3 axioms:

1. $P(S) = 1$

2. $P(A) \geq 0$, $\forall A \subset S$

3. $P(\cup A_k) = \sum P(A_k)$

where A_k mutually exclusive

→ Mutually exclusive: $A_i \cap A_j = \emptyset$, $\forall i, j$

→ collectively exhaustive: $\cup A_k = S$

→ Random variable is a function whose domain is the set of outcomes $\omega \in S$ and whose range is the real line. For every outcome $\omega \in S$ we assign a number $X(\omega)$ s.t. $\{\omega: X(\omega) \leq x\}$ is an event and

$$P(\{\omega: X(\omega) = \infty\}) = 0 = P(\{\omega: X(\omega) = -\infty\})$$

$$P(X = \infty) = P(X = -\infty) = 0$$

Discrete RVs

PMF: $P_X(x) = P(X=x)$

COF: $F_X(x) = P(X \leq x)$

PDF: $f_X(x) = P(X \text{ is near } x)$

COF: $F_X(x) = P(X \leq x)$

$$f_X(x) = \frac{\partial F_X(x)}{\partial x}$$

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(u) du$$

Expected value: $E(X) = \int_{-\infty}^{\infty} x f_X(x) dx$ $E(g(x)) = \int_{-\infty}^{\infty} g(x) f_X(x) dx$

* $E(X+Y) = E(X) + E(Y)$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x+y) f_{X,Y}(x,y) dx dy = \int_{-\infty}^{\infty} x \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy dx = \int_{-\infty}^{\infty} x f_X(x) dx$$

$$+ \int_{-\infty}^{\infty} y \int_{-\infty}^{\infty} f_{xy}(x, y) dx dy + \int_{-\infty}^{\infty} y f_y(y) dy = E(x) + E(y)$$

$$\begin{aligned} \sigma_x^2 &= E((X - E(X))^2) = E(X^2) + E(-2E(X)X) + E(E(X)^2) \\ &= E(X^2) - E(X)^2 \end{aligned}$$

→ Expected value is a linear operator

Chapter 7: Parameter Estimation Using the Sample Mean

In previous chapter, we had access to the probability model of a random experiment. We do not assume this anymore

7.1 Sample Mean: Expected value and Variance

Consider repeated independent trials of an experiment.

Each trial results in one observation of an RV, X . After n trials we have sample values of n RVs, $x_1, \dots, x_n$ all with the same PDF as X .

Definition 1: For iid. RVs $x_1, \dots, x_n$ with PDF $f_X(x)$, the sample mean of X RV defined as:

$$M_n(X) = \frac{x_1 + \dots + x_n}{n} \Rightarrow \text{function RVs is a RV itself}$$

$\star \quad M_n(X) \neq E(X) \quad \star$ sample mean (RV) expected value (numbers)

Theorem 1:

$$E(M_n(X)) = E(X)$$

$$\text{Var}(M_n(X)) = \frac{\text{Var}(X)}{n}$$

$E(x_i) = E(X)$ since iid

Proof: $E(M_n(X)) = E\left(\frac{X_1 + \dots + X_n}{n}\right) = \frac{1}{n} [E(X_1) + \dots + E(X_n)] = \frac{n E(X)}{n} = E(X)$

$$\text{Var}(M_n(X)) = \text{Var}\left(\frac{X_1 + \dots + X_n}{n}\right) = \frac{1}{n^2} \text{Var}(X_1 + X_2 + \dots + X_n) = \frac{1}{n^2} n \text{Var}(X) = \frac{\text{Var}(X)}{n}$$

$$\begin{aligned} * \text{Var}(X+Y) &= E[(X+Y - (\mu_X + \mu_Y))^2] = E[X^2 - \mu_X^2 + Y^2 - \mu_Y^2 + 2(XY - \mu_X \mu_Y)] = \\ &= \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y) \end{aligned}$$

As $n \rightarrow \infty$, $M_n(X) \rightarrow \infty$
 $\hookrightarrow$ sample mean converges to expected value.

4.2 Derivation of a RV from the Expected Value

In order to understand the convergence of $M_n(X)$ to $E(X)$, we start the analysis with bounding

Theorem 2: Markov Inequality

For a RV X s.t. $P(X < 0) = 0$ and a constant c .

$$P(X \geq c^2) \leq \frac{E(X)}{c^2}$$

Pf: X is nonnegative $\Rightarrow f_X(x) = 0, x < 0$

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} x f_X(x) dx = \int_0^{\infty} x f_X(x) dx = \int_0^{c^2} x f_X(x) dx + \int_{c^2}^{\infty} x f_X(x) dx \\ &\geq \int_{c^2}^{\infty} x f_X(x) dx \geq \int_{c^2}^{\infty} c^2 f_X(x) dx = c^2 \int_{c^2}^{\infty} f_X(x) dx \end{aligned}$$

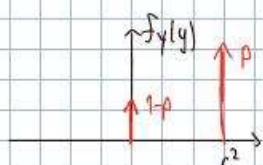
$$P(X \geq c^2) \leq \frac{E(X)}{c^2}$$

Example 1: Let X be the height of a randomly chosen adults. If the expected value of X is 160 cm, then the prob X being smaller than 320 cm is

$$P(X \geq 320) \leq \frac{160}{320} = \frac{1}{2}$$

$\Rightarrow$ Markov inequality is very loose for many RVs

Example 2: Consider RV. Y that takes on the value c^2 with prob p and 0 with $1-p$



$$E(Y) = 0(1-p) + c^2 p = c^2 p$$

$$P[Y \geq c^2] \leq \frac{E(Y)}{c^2} = \frac{c^2 p}{c^2} = p$$

We know that $P(Y \geq c^2) = P(Y = c^2) = p \Rightarrow P[Y \geq c^2] = \frac{E(Y)}{c^2}$
 Markov Inequality becomes equality

Chebyshev Inequality

$\rightarrow$ Sinus sorusu: Choose such n , to keep error in some bound?
 derived RV

Theorem 3: Define RV $X = (Y - \mu_Y)^2$. For an arbitrary constant $c > 0$,

$$P[|Y - \mu_Y| \geq c] \leq \frac{\text{Var}(Y)}{c^2}$$

Pf. Apply M.I. to RV X

$$P(X \geq c^2) \leq \frac{E(X)}{c^2}$$

$$P[|Y - \mu_Y| \geq c] = P[(Y - \mu_Y)^2 \geq c^2] \leq \frac{E[(Y - \mu_Y)^2]}{c^2} = \frac{\text{Var}(Y)}{c^2}$$

$\rightarrow$ Unlike M.I, C.I. is valid for all random RVs.

$\rightarrow$ Generally provides a tighter bound.

$\rightarrow$ When $\text{Var}(Y)$ is small, C.I. says that it is unlikely that Y is far from its value.

Example 3: Following Ex1, $E(X) = 160\text{cm}$ and $\sigma_X = 30\text{cm}$, Calculate the C.I. to find an upper-bound on $P(X \geq 320)$

$$P(X \geq 320) = P(X - 160 \geq 320 - 160)$$

$$P(|X - \mu_X| \geq 160) \leq \frac{30^2}{160^2} = 0.035$$

→ Still loose but much tighter than M.I.

7.3 Point Estimates of Model Parameter

We try to estimate a "parameter" of a probabilistic model.

↳ any number that can be calculated from the model

For example, for a given experiment assume that we want to estimate the prob. of an event A , $P[A]$.

⇒ $P[A]$ is a model parameter. We try to define a new R.V. whose exp. value $E[X] = P[A]$

If we can define such a R.V., we can simply utilize the sample mean, to estimate $P[A]$
 $M_n(X)$

Let us define an indicator RV X_A

$$X_A = \begin{cases} 1 & \text{if event } A \text{ occurs} \\ 0 & \text{if event } A \text{ doesn't occur} \end{cases}$$

$$\Rightarrow E[X_A] = 1 \cdot P[A] + 0 \cdot P[\bar{A}] = P[A]$$

⇒ $M_n(X_A)$ can be used to estimate $E[X_A] = P[A]$

$$\Rightarrow \hat{P}_n(A) = M_n(X_A) = \frac{X_{A_1} + X_{A_2} + \dots + X_{A_n}}{n}$$

↳ relative freq. of event A in n trials.

Two types of estimators:

→ Point estimate: Single number as close as possible to the parameter to be estimated

→ Confidence interval: Range of numbers that contain the parameter to be estimated with high probability.

Properties of point estimators $\begin{cases} \text{bias} \\ \text{consistency} \\ \text{accuracy} \end{cases}$

Setup: We want to estimate a parameter r of the prob. model of R.V
Let $X_1, X_2, \dots, X_n$ all have the same prob model as X . We use the observations of $X_1, X_2, \dots, X_n$ to produce a family of estimators of r .

- $\rightarrow \hat{R}_1$ is an estimator of r using X_1
- $\rightarrow \hat{R}_2$ is an estimator of r using X_1 and X_2
- $\vdots$
- $\rightarrow \hat{R}_n$ is an estimator of r using $X_1, \dots, X_n$

X $\xRightarrow{\text{samples}}$ $X_1, X_2, X_3, \dots, X_n$
 $\downarrow$ $\hat{R}_3$: estimate of r using 3 observations.
parameter r

Definition 2: The sequence of estimators $\hat{R}_1, \hat{R}_2, \dots$ of parameter r is consistent if for any $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} P[|\hat{R}_n - r| \geq \epsilon] = 0$$

Definition 3: An estimate $\hat{R}$, of parameter r is unbiased if $E[\hat{R}] = r$, otherwise $\hat{R}$ is biased.

Definition 4: The seq. of estimators $\hat{R}_n$ of parameter r is asymptotically unbiased if $\lim_{n \rightarrow \infty} E[\hat{R}_n] = r$

Definition 5: The mean square error (MSE) of estimator $\hat{R}$ of param. r is

$$e = E[(\hat{R} - r)^2]$$

Very Important Observation: When $\hat{R}$ is an unbiased estimator of r ($E[\hat{R}] = r$) then the mean square error e is the variance of $\hat{R}$.
If you can show that the MSE goes to zero, it shows that the estimator is consistent.

Theorem 4: If a seq. of unbiased estimators $\hat{R}_1, \hat{R}_2, \dots, \hat{R}_n$ of parameter r has MSE $e_n = \text{Var}[\hat{R}_n]$ satisfying $\lim_{n \rightarrow \infty} e_n = 0$, then the seq. is consistent.

Pr: Since $E[\hat{R}_n] = r$, we can apply C.I. as

$$P[|\hat{R}_n - r| \geq \epsilon] \leq \frac{\text{Var}[\hat{R}_n]}{\epsilon^2}$$

$$\lim_{n \rightarrow \infty} P[|\hat{R}_n - r| \geq \epsilon] \leq \lim_{n \rightarrow \infty} \frac{\text{Var}[\hat{R}_n]}{\epsilon^2} \stackrel{e_n \rightarrow 0}{=} 0$$

$$\Rightarrow \lim_{n \rightarrow \infty} P[|\hat{R}_n - r| \geq \epsilon] = 0 \quad \text{since } P[\cdot] \geq 0$$

Example 4: In any interval of k seconds the number of packets, N_k , that pass through an internet router has Poisson distribution with $E(N_k) = k \cdot r$. Assume that we want to estimate the parameter r using $\hat{R}_k = \frac{N_k}{k}$. Is this an unbiased estimate of r ? What is the MSE, e_k ? Is the seq. of estimators $\hat{R}_1, \dots, \hat{R}_n$ consistent?

$$E[\hat{R}_k] = E\left[\frac{N_k}{k}\right] = \frac{1}{k} E[N_k] \stackrel{k \cdot r}{=} r \Rightarrow \text{unbiased}$$

For Poisson R.V., $E[N_k] = \text{Var}[N_k] = k \cdot r$

$$e_k = \text{Var}[\hat{R}_k] = \text{Var}\left[\frac{N_k}{k}\right] = \frac{1}{k^2} \text{Var}[N_k] = \frac{1}{k^2} k \cdot r = \frac{r}{k}$$

$$\lim_{k \rightarrow \infty} e_k = 0 \Rightarrow \text{consistent} \quad \left(\begin{array}{l} \text{since unbiased} \\ \text{according to Thm 4} \end{array} \right)$$

Point Estimates of the Expected Value

To estimate $\mu = E[X]$, we use the sample mean $\hat{R}_n = M_n(X)$

Theorem 5: The sample mean is an unbiased estimate.

$$\text{Pf: } E[M_n(X)] = E\left[\frac{X_1 + \dots + X_n}{n}\right] = E[X]$$

Theorem 6: Sample mean estimator has $M_n(X)$ has MSE

$$e_n = E[(M_n(X) - E(X))^2] = \text{Var}[M_n(X)] = \frac{\text{Var}(X)}{n}$$

e_n : variance of the sample mean

$\sqrt{e_n}$: st. deviation - the sample mean $\rightarrow$ also called the standard error

$\rightarrow \sqrt{e_n}$ gives us an idea about how far we should expect the sample mean to deviate from the expected value

If X is a Gaussian RV,

$$P[E(X) - \sqrt{e_n} \leq M_n(X) \leq E(X) + \sqrt{e_n}] = 2\Phi(1) - 1 = 0.68$$

Example 6: How many indep. trials, n , are needed to guarantee that $\hat{P}_n(A)$, the relative freq. estimate of $P(A)$, has standard error less than 0.01?

$$\hat{P}_n(A) = M_n(X_A)$$

$$\text{Var}(\hat{P}_n(A)) = \text{Var}(M_n(X_A)) = \frac{\text{Var}(X_A)}{n}$$

$$\text{where } X_A = \begin{cases} 1, & A \text{ occurs} \\ 0, & \text{o.w.} \end{cases}$$

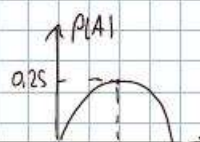
$$\text{Var}(X_A) = \underbrace{E[X_A^2]} - E[X_A]^2 = P(A) - P(A)^2 = P(A)(1 - P(A))$$

$$E[X_A^2] = E[X_A] = P(A)$$

$$\text{Var}[\hat{P}_n(A)] = \frac{P(A)[1 - P(A)]}{n} = e_n$$

$$\rightarrow \text{we want } \frac{P(A)[1 - P(A)]}{n} \leq 0.01$$

$$\text{For } 0 \leq P \leq 1, P(1-P) \leq 0.25 \Rightarrow$$



We need at least $n=25$

0 0.5 1 'A

Theorem 7: If X has finite variance, then the sample mean $M_n(X)$ is a seq. of consistent estimators of $E(X)$

Pf: $\lim_{n \rightarrow \infty} \text{Var}[M_n(X)] = \lim_{n \rightarrow \infty} \frac{\text{Var}(X)}{n} = 0$

$\rightarrow \text{finite}$
 $\rightarrow \text{inf}$

This result is known as the weak law of large numbers.

Theorem 8: If X has finite variance, for any constant $c > 0$,

a) $\lim_{n \rightarrow \infty} P[|M_n(X) - \mu_X| \geq c] = 0$

b) $\lim_{n \rightarrow \infty} P[|M_n(X) - \mu_X| < c] = 1$

Remember that we've defined the relative freq. of event A as

$\hat{P}_n(A) = M_n(X_A)$ whose mean is $P(A)$. We can now state the same thing using the weak law of large num.s.

Theorem 9: As $n \rightarrow \infty$, $\hat{P}_n(A)$ converges for $P(A)$; i.e., for any $c > 0$

$$\lim_{n \rightarrow \infty} P[|\hat{P}_n(A) - P(A)| \geq c] = 0$$

When we say weak L.L.N., weak means the way $M_n(X)$ converges to $E(X)$

Defn 6: Convergence in probability: The random sequence Y_n converges in probability to a constant y if for any ϵ

$$\lim_{n \rightarrow \infty} P[|Y_n - y| \geq \epsilon] = 0$$

When we say strong L.L.N., we don't deal with probabilities, we mean the seq. $M_1, M_2, \dots$ converges to M_X

For practical applications, difference between weak and strong law is not important.

Strong Law $\Rightarrow$ Convergence almost everywhere
 " " " surely always

Point Estimators of the Variance

Next task is estimating $\mu = \text{Var}(X)$

$$\text{Var}(X) = E[(X - \mu_X)^2] \rightarrow \mu_X = E(X) \text{ is known}$$

Suppose we know that $\mu_x = 0$

$\Rightarrow \text{Var}(X) = E(X^2)$. We can simply define $Y = X^2 \Rightarrow \text{Var}(X) = E(Y)$
and we know that $M_n(Y)$ is a good enough estimator for the exp. value

$$M_n(Y) = \frac{Y_1 + Y_2 + \dots + Y_n}{n} = \frac{X_1^2 + X_2^2 + \dots + X_n^2}{n}$$

→ Now assume M_x is known but nonzero.

$\Rightarrow$ We can define $W = (X - \mu_X)^2$ which leads to

$$M_n(W) = \frac{W_1 + W_2 + \dots + W_n}{n} = \frac{1}{n} \sum_{i=1}^n (X_i - \mu_X)^2$$

If $\text{Var}(W)$ exists, $M_n(W)$ is a consistent unbiased estimate of $\text{Var}(X)$.

★ If μ_X is unknown, we're in trouble, we need to estimate it using $M_n(X)$

$$V_n(X) = \frac{1}{n} \sum_{i=1}^n (X_i - M_n(X))^2$$

Theorem 10: In contrast to the sample mean, the sample variance is a biased estimator of $\text{Var}(X)$

pp.

$$V_n(X) = \frac{1}{n} \sum_{i=1}^n (X_i - M_n(X))^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - M_n^2(X) \quad \begin{array}{l} \text{since} \\ E[(X - \mu_X)^2] \\ = E[X^2] - \mu_X^2 \end{array}$$

$$= \frac{1}{n} \sum_{i=1}^n X_i^2 - \left(\frac{X_1 + X_2 + \dots + X_n}{n} \right) \left(\frac{X_1 + X_2 + \dots + X_n}{n} \right)$$

$$= \frac{1}{n} \sum_{i=1}^n X_i^2 - \frac{1}{n^2} \sum_i \sum_j X_i X_j$$

$$E(V_n(X)) = \frac{1}{n} \left[\overbrace{E(X^2) + \dots + E(X^2)}^{n \text{ add}} \right]$$

$$- \frac{1}{n^2} \sum_i \sum_j (Cov(X_i, X_j) + \mu_X^2)$$

$$= E(X^2) - \frac{1}{n^2} (n^2 \mu_X^2 + n \text{Var}(X))$$

$$= E(X^2) - \mu_X^2 - \frac{1}{n} \text{Var}(X) = \left(\frac{n-1}{n} \right) \text{Var}(X) \Rightarrow \text{BIASED estimator}$$

$$E[X_i, X_j] = E[X_i]E[X_j] + Cov[X_i, X_j]$$

$$Cov(X_i, X_j) = \begin{cases} \text{Var}(X), & i=j \\ 0, & i \neq j \end{cases}$$

Since X_i are independent

Even though $V_n(X)$ is a biased estimator, it's asymptotically unbiased, i.e.,

$$\lim_{n \rightarrow \infty} E[V_n(X)] = \lim_{n \rightarrow \infty} \frac{n-1}{n} \text{Var}(X) = \text{Var}(X)$$

Theorem 11.

The estimate $V_n'(X) = \frac{1}{n-1} \sum_{i=1}^n (X_i - M_n(X))^2$ is an unbiased estimator $\text{Var}(X)$

Pf

$$V_n'(X) = \frac{n}{n-1} V_n(X)$$

$$\hookrightarrow E(V_n'(X)) = \frac{n}{n-1} E(V_n(X)) = \text{Var}(X)$$

7.4 Confidence Intervals

Asymptotically most estimators seem fine. In practice, we will need to assess the accuracy for a finite num of samples.

Convergence of the Sample Mean to the exp value.

We will apply to C.I. to $Y = M_n(X)$

Theorem 12:

For any constant $c > 0$,

$$a) \quad P[|M_n(X) - \mu_X| \geq c] \leq \frac{\overbrace{\text{Var}[M_n(X)]}^{\text{Var}(X)}}{\underbrace{n}_{n} \underbrace{c^2}_{c^2}} = \alpha$$

$$b) \quad P[|M_n(X) - \mu_X| < c] \geq 1 - \frac{\text{Var}(X)}{n c^2} = 1 - \alpha$$

Pf:

$$E(M_n(X)) = E(X), \quad \text{Var}[M_n(X)] = \frac{\text{Var}(X)}{n}$$

We will simply apply C.I.

This theorem dealt with interval: $|M_n(X) - \mu_X| < c$

$$\Rightarrow -c < M_n(X) - \mu_X < c$$

$$\Rightarrow \mu_x - c < M_n(X) < \mu_x + c$$

↳ This $2c$ -long interval is called confidence interval.

The prob that the RV, $M_n(X)$ is inside this interval is $1-\alpha$, which is the confidence interval

Example: Suppose we perform n ind. trial of an experiment and use $\hat{P}_n(A)$ to estimate $P(A)$.

Use the CI. to calculate the smallest n such that $\hat{P}_n(A)$ is in a confidence interval of length 0.02 with confidence coefficient 0.999

Remember that $\hat{P}(A)$ is the sample mean of RV X_A .

$$E(X_A) = P(A) \quad , \quad \text{Var}(X_A) = P(A) \cdot (1-P(A))$$

$$\Rightarrow \Pr[|X - E(X)| \geq c] \leq \frac{\text{Var}(X)}{c^2} \quad \text{also} \quad \hat{P}_n(A) = M_n(X_A)$$

$$\Pr[|\hat{P}_n(A) - P(A)| \geq c] \leq \frac{P(A)(1-P(A))}{nc^2}$$

$$\Pr[|\hat{P}_n(A) - P(A)| < c] \geq 1 - \frac{P(A)(1-P(A))}{nc^2} \geq 1 - \frac{1}{4nc^2} \geq 0.999$$

$$2c = 0.02 \rightarrow c = 0.01$$

For $0 \leq p \leq 1$, $p(1-p) \leq 0.25$

$$1 - \frac{1}{4nc^2} \geq 0.999 \Rightarrow \frac{1}{4n(0.01)^2} \leq 0.001$$

$$\Rightarrow n \geq 2.5 \times 10^6 \text{ trials}$$

0.999

In the case of a rare event ($P(A)$ close to zero), we expect to perform many more trials to estimate $P(A)$

Ex 7

In a similar setup, we want the relative frequency $\hat{P}_n(A)$ to differ by less than 10%

$$P[|\hat{P}_n(A) - P(A)| \geq \underbrace{0.1 P(A)}_c] \leq \frac{P(A)(1-P(A))}{n \times 0.01 \times P(A)^2}$$

$$= \frac{1 - P(A)}{n \times 0.01 \times P(A)} \leq \frac{1}{n \times 0.01 \times P(A)} = \frac{100}{n \times 0.01}$$

$$\frac{100}{n P(A)} \leq 0.001 \Rightarrow n \geq \frac{100\,000}{P(A)}$$

Interval Estimates of model Parameters

In confidence interval estimation, we consider a range of parameters and a probability that the parameter is in the selected range:

$$P[A \leq r \leq B] \geq 1 - \alpha$$

$B-A$: Confidence interval, A & B may be random

$1-\alpha$: Confidence coeff

If the parameter of interest is $r = E(X)$ of a RV X ,

$$P[M_n(X) - c < E(X) < M_n(X) + c] \geq 1 - \frac{\text{Var}(X)}{nc^2}$$

Example 8: Suppose X , is the ind. measurement of a board whose actual length is b cm.

$\Rightarrow$ Each X_i has the form $X_i = b + Z_i$, $\mu_z = 0 \text{ cm}$
 $\sigma_z = 1 \text{ cm}$

How many measurements are needed to achieve a conf. interval of length $2c = 0.2$ with $1 - \alpha = 0.99$?

$$E(X_i) = b + E(Z_i) = b, \text{Var}(X_i) = \text{Var}(Z_i) \Rightarrow b \text{ not probabilistic}$$

$$P[|M_n(X) - b| < 0.1] \geq 1 - \frac{1}{n(0.1)^2}$$

$$\Rightarrow P[M_n(X) - 0.1 < b < M_n(X) + 0.1] \geq 1 - \frac{100}{n} \geq 0.99$$

$$\frac{100}{n} \leq 0.01 \Rightarrow n = 10\,000 \text{ measurements}$$

In practice, we can probably achieve this interval with fewer measurements, but cannot model that without more information about the prob. model of Z_i .

Notice that $M_n(X)$ is a summation of many many RVs

$\Rightarrow$ Central limit theorem is triggered.

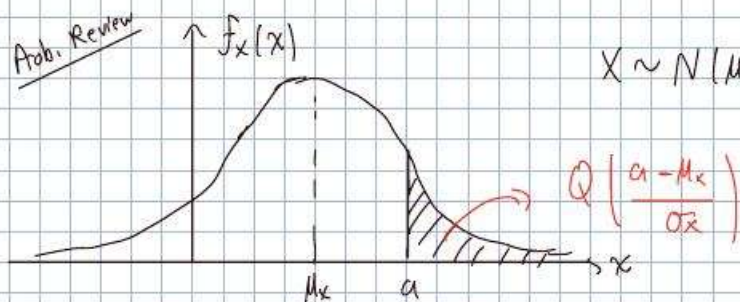
Theorem 13: Let $X \sim N(\mu_x, \sigma_x)$. A confidence interval estimate of μ_x of the form

$$M_n(X) - c \leq \mu_x \leq M_n(X) + c$$

has confidence coeff. $1 - \alpha$ with

$$\frac{\alpha}{2} = Q\left(\frac{c\sqrt{n}}{\sigma_x}\right) = 1 - \Phi\left(\frac{c\sqrt{n}}{\sigma_x}\right)$$

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$$X \sim N(\mu_X, \sigma_X) \Rightarrow f_X(x) = \frac{1}{\sqrt{2\pi\sigma_X^2}} e^{-\frac{(x-\mu_X)^2}{2\sigma_X^2}}$$

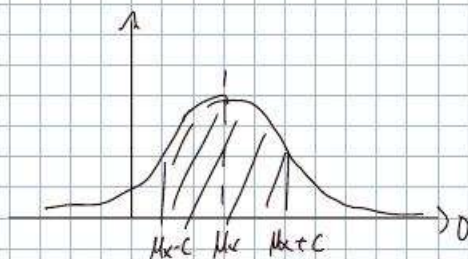
PF

$$P[M_n(X) - c \leq \mu_X \leq M_n(X) + c]$$

$$= P[\mu_X - c \leq M_n(X) \leq \mu_X + c]$$

$$= 1 - 2Q\left(\frac{c}{\sigma_{M_n(X)}}\right)$$

$$= 1 - 2Q\left(\frac{c\sqrt{n}}{\sigma_X}\right)$$



This especially holds when X is Gaussian. Even if X is not Gaussian, $M_n(X)$ can still be a Gaussian due to CLT.

Example 9: Repeat ex 8 with the knowledge that Z_i are iid Gaussian. We again have

$$M_n(X) - 0.1 < b < M_n(X) + 0.1$$

Since $X_i \sim N(b, 1)$

$$\alpha = 2Q\left(\frac{0.1\sqrt{n}}{1}\right) \leq 0.01$$

$$\Rightarrow n \geq 666$$

$\Rightarrow$ Without the prob model, we determined $n = 10k$ as the min. number.

Example 10: $Y \sim N(\mu_Y, \sigma_Y)$. Use $M_n(Y)$ to find a confidence interval of estimate of μ_Y with confidence 0.99. If $\sigma_Y^2 = 10$ and $M_{100}(Y) = 33.2$, what's our interval estimate of μ_Y from 100 samples?

→ For $1 - \alpha = 0.99$, Theorem 13 states

$$P(M_n(Y) - c \leq \mu_Y \leq M_n(Y) + c) = 1 - \alpha = 0.99$$

$$\frac{\alpha}{2} = 0.005 = 1 - \Phi\left(\frac{c\sqrt{n}}{\sigma_Y}\right)$$

Using the CDF/Q table, $\frac{c\sqrt{n}}{\sigma_Y} = 2.58 \Rightarrow c = \frac{2.58 \sigma_Y}{\sqrt{n}}$

$$\Rightarrow M_n(Y) - \frac{2.58 \sigma_Y}{\sqrt{n}} \leq \mu_Y \leq M_n(Y) + \frac{2.58 \sigma_Y}{\sqrt{n}}$$

for the given values, $32.384 \leq \mu_Y \leq 34.016$

* ISAC, JSAC acoustic, Passive radar, Nonlinear Junction Detection

Chapter 8: Hypothesis Testing

We analyze observations of an experiment to arrive at a conclusion. When the conclusion is based on RVs, the reason is called statistical inference.

↳ Point estimation

↳ Conf. Interval Estimation

↳ Significance testing: accept or reject the hypothesis that observations results from a certain prob. model: H_0

↳ Hypothesis testing: Observations result from one of M hypothetical prob. models: $H_0, H_1, \dots, H_{M-1}$

8.1 Significance Testing

H_0 : null hypothesis $\rightarrow$ accept H_0
 $\rightarrow$ reject H_0

The significance level of the test is defined as the prob. of rejecting H_0 if it's true

S_0 : Sample space $\Rightarrow$ We split it into $S = \underbrace{A}_{\text{accept}} \cup \underbrace{R}_{\text{reject}}$
 $\Rightarrow R = A^c$

We accept H_0 if $s \in A$, reject if $s \in R$

$\Rightarrow$ Significance level $\alpha = P[s \in R]$ when H_0 true.

To design a significance test we first pick α and then determine the set R that satisfies the prob. condition.

In many applications H_0 , the prob. model does not change.

Example 11:

Assume that number of tweets, N , at 9.00 and 9.30 on Thursday is Poisson RV with mean 1000. Next Thursday, a president candidate will give a speech at 9.00. Will that change the prob. model? Design a test for H_0 at $\alpha = 0.05$

H_0 : $N \sim \text{Poisson}$ with mean 1000 on the night of the speech.

We need to define R s.t. $P[N \in R] = 0.05$

$\rightarrow$ The speech can increase or decrease the number of tweets:

$$R = \{n : |n - 1000| \geq c\}$$

Given this detection scheme, find c .

* However, there are many possible R sets.

$$\Rightarrow \alpha = P[|N - 1000| \geq c] \Rightarrow N \sim \text{Poisson}(1000)$$

$$= P\left[\frac{|N - E[N]|}{\sigma_N} \geq \frac{c}{\sigma_N}\right] \rightarrow \begin{array}{l} \text{For Poisson RV} \\ \text{Var}[N] = E[N] \\ \sigma_N = \sqrt{1000} \end{array}$$

Since $E[N]$ is large, CLT causes $\frac{N - E[N]}{\sigma_N}$ to converge to $Z \sim N(0,1)$

$$\Rightarrow \alpha = P\left[|Z| \geq \frac{c}{\sqrt{1000}}\right] = 2Q\left(\frac{c}{\sqrt{1000}}\right) = 0.05$$

$$\Rightarrow c = 61.7$$

We reject the null hypothesis if $N \leq 938$ and $N \geq 1062$ tweets

Type-I error: H_0 true, reject H_0 : false reject

Type-II error: H_0 false, accept H_0 : false acceptance

Prob. of type-I error is simply $\alpha = P[s \in R]$ when H_0 is true

Prob. of type-II error is not easy since we don't know the prob. model. We will need to define H_1 .

Example 12 : We are testing a diet pill on 64 people. Before the test, the prob. model for the people is a RV, w

$$W \sim N(190, 24) \text{ lbs}$$

Design a test based on sample mean of the weight of the population to determine whether the pill had a "significant" effect. The significance level is $\alpha = 0.01$.

H_0 : The prob. model remains as $W \sim N(\overset{\mu}{190}, \overset{\sigma^2}{24})$

$M_{64}(W)$, the sample mean is distributed as $\sim N(190, \underbrace{\frac{24}{64}}_{=3})$

$$\text{Var}(M_n(X)) = \frac{\text{Var}(X)}{n}$$

$\Rightarrow$ Significance test : $P[M_{64}(W) \in R] = \alpha = 0.01$

accept H_0 : diet pills don't have a significant effect

reject H_0 : diet pill works

$$\Rightarrow R = \{ M_{64}(W) \leq r_0 \}$$

$$P[M_{64}(W) \leq r_0] = \Phi\left(\frac{r_0 - 190}{3}\right) = 0.01$$

$$\Rightarrow r_0 = 183.01,$$

$\rightarrow$ If the ave weight is less than 183.01, we reject H_0 and state that diet pill works.

$\rightarrow$ Ex 11 had symmetric R , Ex 12 had one-sided R .

8.2 Binary Hypothesis Testing

We now have 2 hypothesis, H_0 or H_1 . We either accept H_0 or H_1 . We either accept H_0 or accept H_1 . We also define a priori probabilities, $P(H_0)$ and $P(H_1)$. They reflect the state of knowledge before we observe an outcome.

H_0 and H_1 , have the same sample space, S . We also define $S' = \{H_0, H_1\}$

→ We observe an outcome $s \in S$. When the observation leads to a random vector $\bar{X}$, we call the $\bar{X}$ the decision statistic. Often, we pick $\bar{X} = X$ (single RV). When X is discrete, we define $P_{X|H_0}(x)$ and $P_{X|H_1}(x)$ or, for continuous X , we define $f_{X|H_0}(x)$ and $f_{X|H_1}(x)$. Conditional prob mass/density function

↙ We also call them likelihood functions

The test divides S into two: $S = A_0 \cup A_1$ where $A_1 = A_0^c$

If $s \in A_0$, accept H_0

If $s \in A_1$, accept H_1

There can be two errors $\Rightarrow$ Type-I error: $P[A_1|H_0]$
Type-II error: $P[A_0|H_1]$

For radar applications, H_0 : no target
 H_1 : target present

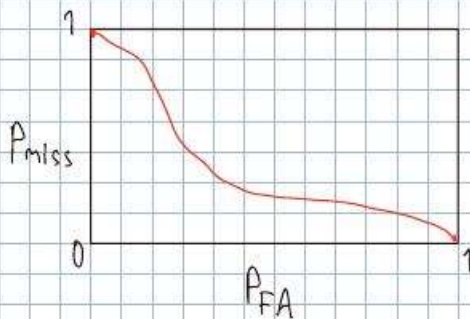
Type-I Error: False Error (FA)

Type-II Error: miss (MISS)

Consider an extreme design $A_0 = S$ and $A_1 = \emptyset$

↳ Then $P_{FA} = P[A_1 | H_0] = 0$, $P_{Miss} = P[A_0 | H_1] = 1$

→ The trade-off can be visualised on a receiver operating curve (ROC).



Depending on the choice of A_0 and A_1 , you can achieve different points on ROC

⇒ You are operating in a point on this curve.

We can also have the prob. models approximately.

$P_{X|H_0}(x)$
 $P_{X|H_1}(x)$ } X : discrete
likelihood function

$f_{X|H_0}(x)$
 $f_{X|H_1}(x)$ } X : cont.
likelihood functions

Example 13: The noise voltage in a radar system is $N \sim N(0, 1)$. When a target is present, the received signal is present, the received signal is $X = V + N$, $V \geq 0$

Assume that acceptance tests are

$A_0 = \{X \leq x_0\}$ and $A_1 = \{X \geq x_0\}$

Obtain ROC of this system for $V = 0, 1, 2$

H_0 : no target, H_1 : target present

↳ We need to obtain P_{FA} and P_{Miss} as a funct of x_0

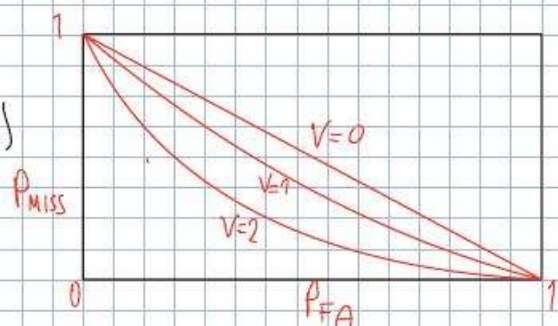
$$P_{FA} = P[\text{accept } H_1 \mid \underline{H_0 \text{ is correct}}] = P[A_1 | H_0] = P[N > x_0]$$

determines the condition

$$P_{\text{Miss}} = P[\text{accept } H_0 \mid \underbrace{H_1 \text{ is correct}}_{\text{determines the received signal}}] = P[V+N \leq x_0]$$

$$P_{\text{FA}} = P[N > x_0] = 1 - \Phi(x_0)$$

$$P_{\text{Miss}} = P[N \leq x_0 - V] = \Phi(x_0 - V)$$



Maximum A Posteriori (MAP) Test

Theorem 14: Given binary hypothesis test with outcome \$s\$, the MAP test leads to the lowest possible \$P_{\text{ERR}}\$ (total prob. of error):

\$s \in A_0\$ if \$P[H_0|s] \geq P[H_1|s]\$ and \$s \in A_1\$ otherwise

Proof:

$$P_{\text{ERR}} = P[A_1|H_0]P[H_0] + P[A_0|H_1]P[H_1]$$

Consider the contribution of \$s\$ in the overall error. We need to place it in \$A_0\$ if \$P[s|H_1]P[H_1] \leq P[s|H_0]P[H_0]\$ and in \$A_1\$ otherwise to minimise the overall error.

$$\Rightarrow s \in A_0 \text{ if } P(s|H_1)P[H_1] \leq P(s|H_0)P[H_0]$$

$$\Leftrightarrow P[s, H_1] \leq P[s, H_0] \Leftrightarrow P[H_1|s] \leq P[H_0|s]$$

$$\underbrace{P[H_1|s]} \leq \underbrace{P[H_0|s]}$$

they represent our a posteriori probabilities \$\Rightarrow\$ knowledge after observations

Theorem 15: For an experiment that produces a random vector $\bar{X}$, the MAP hypothesis test is:

Discrete: $\bar{x} \in A_0$, if $\frac{P_{\bar{x}|H_0}(\bar{x})}{P_{\bar{x}|H_1}(\bar{x})} \geq \frac{P[H_1]}{P[H_0]}$; $\bar{x} \in A_1$ otherwise

Continuous: $\bar{x} \in A_0$, $\frac{f_{\bar{x}|H_0}(\bar{x})}{f_{\bar{x}|H_1}(\bar{x})} \geq \frac{P[H_1]}{P[H_0]}$

Example 14: For a digital communication system, we transmit 0 with prob. p , 1 with prob. $1-p$.

$$\begin{array}{l} 0: X = -V + N \\ 1: X = V + N \end{array} \left. \begin{array}{l} \text{Received signal } \pm V: \text{ deterministic signal comp.} \\ N: N(0, \sigma) \text{ noise component} \end{array} \right\}$$

Given the received signal X , what is the min prob of error rule for deciding whether 0 or 1 was transmitted.

$$H_0: 0 \text{ transmitted} \Rightarrow X \sim N(-v, \sigma)$$

$$H_1: 1 \text{ transmitted} \Rightarrow X \sim N(+v, \sigma)$$

$$\Rightarrow f_{X|H_0}(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x+v)^2/2\sigma^2}$$

$$f_{X|H_1}(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-v)^2/2\sigma^2}$$

$$\Rightarrow x \in A_0 \text{ if } \frac{e^{-(x+v)^2/2\sigma^2}}{e^{-(x-v)^2/2\sigma^2}} \geq \frac{1-p}{p}, \quad x \in A_1 \text{ otherwise}$$

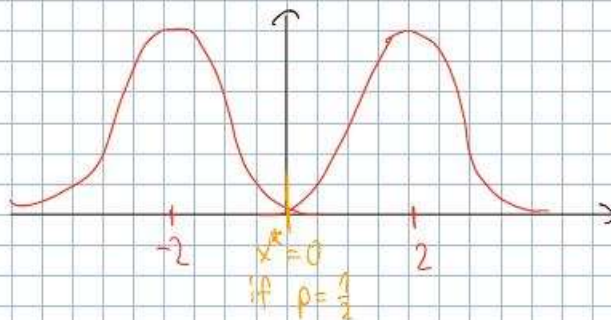
$$x \in A_0 \text{ if } e^{\left[-\frac{(x+v)^2}{2\sigma^2} + \frac{(x-v)^2}{2\sigma^2}\right]} \geq \frac{1-p}{p}$$

$$\Leftrightarrow x \in A_0 \text{ if } e^{-\frac{4xv}{2\sigma^2}} \geq \frac{1-p}{p}$$

$$\Leftrightarrow x \in A_0 \text{ if } -\frac{4xv}{2\sigma^2} \geq \log\left(\frac{1-p}{p}\right) \Rightarrow \text{valid transform since log is monotonous function}$$

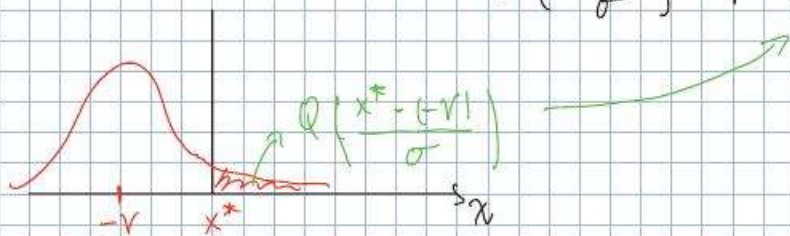
$$\Leftrightarrow x \leq \frac{\sigma^2}{2v} \log\left(\frac{p}{1-p}\right)$$

When $p=1/2$, $x^*=0 \Rightarrow$ we decide H_0 if $x \leq 0$



Example 15: Find the prob of error of the system in Ex 14

$$\begin{aligned} P_{ERR} &= P[A_0|H_1] P[H_1] + P[A_1|H_0] P[H_0] \\ &= P[X \leq x^*] (1-p) + P[X > x^*] p \\ &= (1-p) \Phi\left(\frac{x^*-v}{\sigma}\right) + p Q\left(\frac{x^*+v}{\sigma}\right) \end{aligned}$$



Minimum Cost Test

MAP penalizes both types of errors equally. In some applications, such as radar, we had different costs assigned to Miss and FA. Consider an application where $C = C_{10}$ is the cost of false alarm (decide H_1 when H_0 is true) and $C = C_{01}$ is the cost of miss

$$\begin{aligned} E[C] &= P_{FA} C_{10} + P_{Miss} C_{01} \\ &= P[A_1|H_0] P[H_0] C_{10} + P[A_0|H_1] P[H_1] C_{01} \end{aligned}$$

Theorem 16: For an experiment that produces a R.V. X , the minimum cost is

Discrete:

$$\bar{x} \in A_0 \text{ if } \frac{P_{\bar{x}|H_0}(\bar{x})}{P_{\bar{x}|H_1}(\bar{x})} > \frac{P[H_1] C_{01}}{P[H_0] C_{10}}$$

→ Note that we don't care about the actual values of C_{01} and C_{10} , we are only interested in their ratio.

Neyman - Pearson Test

MAP minimizes the prob. of accepting the wrong hypothesis. The minimum cost test minimizes the cost of errors. They need the knowledge of $P(H_0), P(H_1), C_{01}, C_{10}$. This information, may not be available to us in some applications.

Theorem 17: Based on the cont decision statistic X , Neyman Pearson test minimizes P_{Miss} given that $P_{FA} = \alpha$ by choosing

$$\bar{x} \in A_0 \quad \text{if} \quad L(\bar{x}) = \frac{f_{\bar{x}|H_0}(\bar{x})}{f_{\bar{x}|H_1}(\bar{x})} \geq \gamma$$

$\bar{x} \in A_1$ otherwise, where γ is chosen so that

$$\int_{L(\bar{x}) < \gamma} f_{\bar{x}|H_0}(\bar{x}) d\bar{x} = \alpha$$

Proof: Review of Lagrange Multipliers method

↳ minimize $f(x)$ given $g(x) = 0$

where derivatives zero

⇒ Define $L(x) = f(x) + \lambda g(x)$. Stationary points of $L(x)$ minimizes $f(x)$.

We need to minimize P_{miss} given $P_{\text{FA}} = \alpha$

$$G = P_{\text{miss}} + \lambda (P_{\text{FA}} - \alpha)$$

$$= \int_{A_0} f_{\bar{x}|H_1}(\bar{x}) d\bar{x} + \lambda \left(\underbrace{\int_{A_1} f_{\bar{x}|H_0}(\bar{x}) d\bar{x}}_{1 - \int_{A_0} f_{\bar{x}|H_0}(\bar{x}) d\bar{x}} - \alpha \right)$$

$$= \int_{A_0} \left[f_{\bar{x}|H_1}(\bar{x}) d\bar{x} - \lambda f_{\bar{x}|H_0}(\bar{x}) d\bar{x} \right] + \lambda (1 - \alpha)$$

* For all λ and α , all $\bar{x}$ satisfying

$$f_{\bar{x}|H_1}(\bar{x}) - \lambda f_{\bar{x}|H_0}(\bar{x}) < 0 \quad \text{will minimize } G$$

Theorem 18: If the decision statistic $\bar{x}$ is discrete

$$\bar{x} \in A_0 \quad \text{if} \quad \frac{P_{\bar{x}|H_0}(\bar{x})}{P_{\bar{x}|H_1}(\bar{x})} \geq \gamma, \quad \bar{x} \in A_1 \quad \text{otherwise}$$

where γ is the largest value such that

$$\sum_{L(\bar{x}) < \gamma} P_{\bar{x}|H_0}(\bar{x}) \leq \alpha$$

Neyman-Pearson that is optimal since it minimizes P_{miss} for a given P_{FA} .

Maximum Likelihood Test (ML)

If we do not know the a priori probabilities $P(H_0)$ and $P(H_1)$, ML is another alternative to MAP

Theorem 19: ML decision rule states

$$\text{Discrete: } \bar{x} \in A_0 \quad \text{if} \quad \frac{P_{\bar{x}|H_0}(\bar{x})}{P_{\bar{x}|H_1}(\bar{x})} \geq 1, \quad \bar{x} \in A_1 \quad \text{o.w.}$$

Continuous: $\bar{x} \in A_0$ if $\frac{f_{\bar{x}|H_0}(x)}{f_{\bar{x}|H_1}(x)} \geq 1$, $\bar{x} \in A_1$ o.w

Example 16: In an optical communication link, the photon detector output is Poisson RV K with either an exp. value of 10 000 (H_0) or 1 000 000 (H_1)

Given that both hypothesis are equally likely, design a MAP hypothesis test.

H_0 : Poisson has mean 10 000

H_1 : " " " 1 000 000

$$P_{K|H_0}(k) = \begin{cases} \frac{10^{4k} e^{-10^4}}{k!} & , k = 0, 1, \dots \\ 0 & , \text{o.w.} \end{cases}$$

$$P_{K|H_1}(k) = \begin{cases} \frac{10^{6k} e^{-10^6}}{k!} & , k = 0, 1, \dots \\ 0 & , \text{o.w.} \end{cases}$$

Since $P(H_0) = P(H_1)$, MAP $\equiv$ ML
 $\Rightarrow K \in A_0$ if $\frac{P_{K|H_0}(x)}{P_{K|H_1}(x)} \geq 1$, $K \in A_1$ o.w.

$$\Rightarrow K \in A_0 \text{ if } \frac{10^{4k} e^{-10^4}}{10^{6k} e^{-10^6}} \geq 1$$

$$\Rightarrow \frac{1}{100^k} e^{10^6 - 10^4} \geq 1 \Rightarrow k \leq k^* = \frac{10^6 - 10^4}{\ln 100} = 214.975$$

$\Rightarrow$ If we receive fewer than k^* photons, we decide on H_0

8.3 Multiple Hypothesis Test

When we have more than two hypotheses, we generalize MAP & ML decision rules.

$$P_{\text{CORRECT}} = \sum_{i=0}^{M-1} P[A_i | H_i] P[H_i]$$

Theorem 20: MAP multiple hypothesis test is given as

$$s \in A_m \text{ if } P[H_m | s] \geq P[H_j | s] \text{ for all } j=0, \dots, M-1$$

or equivalently

$$x_i \in A_m \text{ if } P_{x|H_m}(x_i) P[H_m] \geq P_{x|H_j}(x_i) P[H_j], \forall j$$

$$f_{x|H_m}(x_i) P[H_m] \geq f_{x|H_j}(x_i) P[H_j], \forall j$$

Theorem 21: ML multiple hypothesis test is given as

$$s \in A_m \text{ if } P[s | H_m] \geq P[s | H_j], \forall j$$

CHAPTER 9: Estimation of a Random Variable

In previous chapters, we used observations of experiments to predict, estimate several parameters of prob models. This chapter deals with predicting the R.V. directly.

Assume that we are predicting RV X by $\hat{X}$ (also a RV), the quality of the estimate is its MSE.

$$e = E[(X - \hat{X})^2]$$

9.1 Optimum Estimation given another RV

We consider 3 types of information for estimating RV X :

- no observation (blind)
- Observation of X
- observation of Y (another RV)

Theorem 22: In the absence of observation, the MSE estimate of RV X is $\hat{X}_B = E[X]$

PF: Assume we estimate X by $\hat{X}_B$

$$\Rightarrow e = E[(X - \hat{X}_B)^2] = E[X^2] - 2\hat{X}_B E[X] + \hat{X}_B^2$$

In order to minimize e ,

$$\frac{\partial e}{\partial \hat{X}_B} = -2E[X] + 2\hat{X}_B = 0 \Rightarrow \hat{X}_B = E[X]$$

$\Rightarrow$ In the observation $\hat{X}_B = E[X]$

$\rightarrow$ Min error is : $e_B^* = E[(X - \hat{X}_B)^2] = E[(X - E[X])^2] = \text{Var}[X]$

Estimation of X given an Event

We conduct an experiment and instead of observing X directly, we learn that $x \in A$. Given A , we define $f_{X|A}(x)$ or $P_{X|A}(x)$

$\Rightarrow$ We need to minimize

$$e_{X|A} = E[(X - \hat{X}_A)^2 | A]$$

Theorem 23: Given the information $x \in A$, the MSE estimate $\hat{X}_A = E[X|A]$

Example 17: The duration T minutes of a phone call is an exponential RV with $E[T] = 3$ mins. If we observe that the call lasted more than 2 minutes, what is the MSE estimate of duration?

$E_x 3.34$

$$f_T(t) = \begin{cases} \frac{1}{3} e^{-t/3}, & t \geq 0 \\ 0, & \text{o.w} \end{cases}$$

$$f_{T|T \geq 2}(t) = \frac{\frac{1}{3} e^{-t/3}}{P(T \geq 2)} \\ \int_2^{\infty} \frac{1}{3} e^{-t/3} dt = e^{-2/3}$$

The call lasted more than 2 mins $\Rightarrow A = \{T \geq 2\}$

$$\hat{t}_A = E[T | T \geq 2] \Rightarrow \text{min. MSE estimate of } T$$

$$f_{T|T \geq 2}(t) = \begin{cases} \frac{1}{3} e^{-(t-2)/3}, & t \geq 2 \\ 0, & \text{o.w} \end{cases}$$

$$E[T | T \geq 2] = \int_2^{\infty} t \cdot \frac{1}{3} e^{-(t-2)/3} dt = 5 \text{ minutes}$$

$$\Rightarrow \hat{t}_B = 3 \text{ minutes} \Rightarrow \hat{t}_A = 5 \text{ mins}$$
$$\frac{1}{3} \left[t^2 e^{-(t-2)/3} \Big|_2^{\infty} + 3 \int_2^{\infty} e^{-(t-2)/3} dt \right] \\ = \frac{1}{3} \left[6 + 3 \cdot 3 \cdot e^{-2/3} \right] = 5$$

MSE Estimation of X given Y

An experiment produces 2 RVs, X and Y . We observe $Y=y$ and want to predict X . We assign a prediction, $\hat{x}_m(y)$ to each observation $Y=y$.

$$e_m = E[X - \hat{x}_m(y) | Y=y]$$

Theorem 24: The MSE estimate of RV X given the observation $Y=y$ is $\hat{x}_m(y) = E[X | Y=y]$

Example 18: Suppose RVs X and Y are independent. What is the MSE estimate of X given Y ?

$$\begin{aligned}\hat{x}_m(y) &= E[X | Y=y] = \int_{-\infty}^{\infty} x f_{X|Y}(x|y) dx \\ &= f_X(x) \text{ since } X \text{ \& } Y \text{ are indep.} \\ &= \int_{-\infty}^{\infty} x f_X(x) dx = E[X] = \hat{x}_0\end{aligned}$$

$\Rightarrow$ When X & Y are indep., observing $Y=y$ is useless in predicting X .

Example 15: $R \sim \text{Uniform}(0,1)$ and given R ; $X \sim \text{Uniform}(0,r)$
Find $\hat{x}_m(r)$

$$f_{X|R}(x|r) = \begin{cases} 1/r, & 0 \leq x \leq r \\ 0, & \text{other} \end{cases}$$

$$\Rightarrow \hat{x}_m(r) = E[X | R=r] = \int_0^r x \frac{1}{r} dx = \frac{r}{2}$$

Quiz 9.1

$$f_{x,y}(x,y) = \begin{cases} 2(y+x) & 0 \leq x \leq y \leq 1 \\ 0 & \text{o.w.} \end{cases}$$

$$1) f_{x|y}(x|y) = \frac{f_{x,y}(x,y)}{f_y(y)} = \begin{cases} \frac{2}{3y^2}(y+x) & 0 \leq x \leq y \\ 0 & \text{o.w.} \end{cases}$$

$$f_y(y) = \int_0^y f_{x,y}(x,y) dx = 2yx + x^2 \Big|_0^y = 3y^2$$

$$2) \hat{X}_M(y) = E[X | Y=y] = \int_0^y x \left(\frac{2}{3y} + \frac{2x}{3y^2} \right) dx = \left[\frac{x^2}{3y} + \frac{2x^3}{9y^2} \right]_0^y \\ = \frac{y}{3} + \frac{2y}{9} = \frac{5y}{9}$$

↳ MMSE estimator of X given Y is $\hat{X}_M(Y) = 5Y/9$

$$3) f_x(x) = \int_x^1 (2y+2x) dy = \left[y^2 + 2xy \right]_x^1 = 1 - x^2 + 2x(1-x) \\ = 1 + 2x - 3x^2$$

$$f_{y|x}(y|x) = \begin{cases} \frac{2(x+y)}{1+2x-3x^2}, & x \leq y \leq 1 \\ 0 & \text{o.w.} \end{cases}$$

$$4) \hat{y}_M(x) = E[Y | X=x] = \int_x^1 \frac{1}{1+2x-3x^2} [2xy + 2y^2] dy = \frac{xy^2 + \frac{2}{3}y^3}{1+2x-3x^2} \Big|_x^1 \\ = \frac{x(1-x^2) + \frac{2}{3}(1-x^3)}{1+2x-3x^2} = \frac{2+3x-5x^3}{3+6x-9x^2}$$

9.2 Linear Estimation of X given Y

In this section we try to estimate using y of the form,

$$\hat{X}_L(y) = ay + b \quad : \text{linear estimation}$$

const

We choose MSE as the accuracy metric again. We choose a and b to minimize:

$$e_L = E[(X - \hat{X}_L(Y))^2]$$

Theorem 25: RVs X and Y have expected values μ_x and μ_y , std. deviations σ_x and σ_y and correlation coeff. $\rho_{x,y}$

where $\rho_{x,y} = \frac{\text{Cov}(X,Y)}{\sigma_x \sigma_y}$.

The optimum linear MSE (LMSE) estimator of X given Y is

$$\hat{X}_L(Y) = a^*Y + b^*$$

and it has the following properties

a)

$$a^* = \frac{\text{Cov}[X,Y]}{\text{Var}[Y]} = \rho_{x,y} \frac{\sigma_x}{\sigma_y}, \quad b^* = \mu_x - a^* \mu_y$$

b) The minimum MSE for linear estimate is

$$e_L^* = E[(X - \hat{X}_L(Y))^2] = \sigma_x^2 (1 - \rho_{x,y}^2)$$

c) The estimation error is uncorrelated with Y

Proof
a)

$$e_L = E[(X - aY - b)^2] = E[X^2] - 2aE[XY] - 2bE[X] + a^2E[Y^2] + 2abE[Y] + b^2$$

$$\frac{\partial e_L}{\partial a} = -2E[XY] + 2aE[Y^2] + 2bE[Y] = 0$$

$$\frac{\partial e_L}{\partial b} = -2E[X] + 2aE[Y] + 2b = 0$$

$$b^* = E[X] - a^*E[Y],$$

$$E[XY] - a^*E[Y^2] = E[X]E[Y] - a^*E^2[Y]$$

$$a^* = \frac{E[XY] - E[X]E[Y]}{E[Y^2] - E^2[Y]} = \frac{\text{Cov}[X,Y]}{\text{Var}[Y]} = \rho_{X,Y} \frac{\sigma_X}{\sigma_Y},$$

b)

$$e_L = E\left[\left(X - \rho_{X,Y} \frac{\sigma_X}{\sigma_Y} Y - \left(\mu_X - \rho_{X,Y} \frac{\sigma_X}{\sigma_Y} \mu_Y\right)\right)^2\right]$$

$$= E\left[\left|(X - \mu_X) - \left(\rho_{X,Y} \frac{\sigma_X}{\sigma_Y} (Y - \mu_Y)\right)\right|^2\right]$$

$$= E[(X - \mu_X)^2] + \rho_{X,Y}^2 \frac{\sigma_X^2}{\sigma_Y^2} E[(Y - \mu_Y)^2] - 2\rho_{X,Y} \frac{\sigma_X}{\sigma_Y} E[(X - \mu_X)(Y - \mu_Y)]$$

$$= \sigma_X^2 + \rho_{X,Y}^2 \sigma_X^2 - 2\rho_{X,Y} \frac{\sigma_X}{\sigma_Y} \rho_{X,Y} \sigma_X \sigma_Y$$

$$= \sigma_X^2 (1 - \rho_{X,Y}^2),$$

$$\begin{aligned} c) \quad E[Y(X - \hat{X}_L(Y))] &= E[XY] - E\left[Y\left(\rho_{X,Y} \frac{\sigma_X}{\sigma_Y} Y + \left(\mu_X - \rho_{X,Y} \frac{\sigma_X}{\sigma_Y} \mu_Y\right)\right)\right] \\ &= E[XY] - \mu_X \mu_Y + \rho_{X,Y} \frac{\sigma_X}{\sigma_Y} E[Y^2] - \rho_{X,Y} \frac{\sigma_X}{\sigma_Y} \mu_Y^2 \end{aligned}$$

$$= \text{Cov}[X, Y] - \underbrace{\rho_{X,Y} \sigma_X \sigma_Y}_{\text{Cov}[X, Y]} = 0, \quad \Rightarrow \text{orthogonality principle of LMSE}$$

Theorem 26:

When X and Y are bivariate Gaussian RVs, the optimum estimator X given Y is the linear estimator.

Pf:

From Th. 4.29, if X & Y are bivariate Gaussian RVs, the cond PDF of Y given X is

$$f_{Y|X}(y|x) = \frac{1}{\tilde{\sigma}_2 \sqrt{2\pi}} e^{-\frac{(y - \tilde{\mu}_2(x))^2}{2\tilde{\sigma}_2^2}}$$

$$\tilde{\mu}_2(x) = \mu_2 + \rho_{X,Y} \frac{\sigma_2}{\sigma_1} (x - \mu_1), \quad \tilde{\sigma}_2^2 = \sigma_2^2 (1 - \rho_{X,Y}^2)$$

But also $\tilde{X}_L(Y) = \rho_{X,Y} \frac{\sigma_X}{\sigma_Y} (Y - \mu_Y) + \mu_X$ from applying linear estimator.

When X and Y are jointly Gaussian $\hat{X}_M(Y) = E[X|Y] = \tilde{\mu}_X(Y)$

$$\hat{X}_M(Y) = \mu_X + \rho_{X,Y} \frac{\sigma_X}{\sigma_Y} (Y - \mu_Y) = \tilde{X}_L(Y)$$

Example 20:

$R \sim \text{Uniform}(0, 1)$. Given R , $X \sim \text{Uniform}(0, r)$. Derive the optimum linear estimator of R given X .

$$f_{X,R}(x, r) = f_{X|R}(x|r) f_R(r) = \begin{cases} 1/r, & 0 \leq x \leq r \leq 1 \\ 0, & \text{o.w.} \end{cases}$$

$$\hat{\mu}_L(x) = \rho_{R,x} \frac{\sigma_R}{\sigma_x} (x - E[X]) + E[R]$$

$$E[R] = \frac{1}{2}, \quad \sigma_R = \sqrt{\frac{1^2 - 0^2}{12}} = \frac{1}{\sqrt{12}}$$

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,R}(x,r) dr = \begin{cases} \int_0^1 \frac{1}{x} dr = -\ln x & , 0 \leq x \leq 1 \\ 0 & , \text{o.w.} \end{cases}$$

$$E[X] = \int_0^1 \underbrace{x}_{\substack{dv \\ u}} \ln x dx = - \left[\frac{x^2 \ln x}{2} \right]_0^1 - \int_0^1 \frac{x^2}{2} \frac{1}{x} dx = \frac{1}{2} \frac{x^2}{2} \Big|_0^1 = \frac{1}{4}$$

$$E[X^2] = - \int_0^1 x^2 \ln x dx = - \left[\frac{x^3 \ln x}{3} \right]_0^1 - \int_0^1 \frac{x^3}{3} \frac{1}{x} dx = \frac{1}{9} x^3 \Big|_0^1 = \frac{1}{9}$$

$$\text{Var}[X] = \frac{1}{9} - \frac{1}{16} = \frac{7}{9 \times 16} \Rightarrow \sigma_x = \frac{\sqrt{7}}{12}$$

$$E[XR] = \int_0^1 \int_0^r x r \frac{1}{r} dx dr = \int_0^1 \frac{r^2}{2} dr = \frac{1}{6}$$

$$\text{Cov}[X,R] = E[XR] - E[X]E[R] = \frac{1}{6} - \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{24}$$

$$\rho_{X,R} = \frac{\frac{1}{24}}{\frac{\sqrt{7}}{12} \cdot \frac{1}{\sqrt{12}}} = \sqrt{\frac{3}{7}}$$

$$\begin{aligned} \hat{\mu}_L(x) &= \sqrt{\frac{3}{7}} \frac{\frac{1}{\sqrt{12}}}{\frac{\sqrt{7}}{12}} \left(x - \frac{1}{4}\right) + \frac{1}{2} = \frac{6}{7} \left(x - \frac{1}{4}\right) + \frac{1}{2} \\ &= \frac{6}{7} x + \frac{2}{7} \end{aligned}$$

9.3 MAP and ML Estimation

MAP and ML estimators are not optimal, i.e., they do not minimize the MSE, but they are conveniently obtained for many applications

Theorem 27: The MAP estimate of RV X given $Y=y$ is defined as

$$\begin{aligned}\hat{X}_{\text{MAP}}(y) &= \arg \max_x f_{X|Y}(x|y) \\ &= \arg \max_x f_{X,Y}(x,y) \\ &= \arg \max_x f_{Y|X}(y|x) f_X(x)\end{aligned}$$

Proof:

Suppose we observe an event $Y=y$. Let H_x denote the hypothesis that $x \leq X \leq x+dx$. Since x is a cont. parameter, we have a continuum of hypotheses H_x . Choosing a $H_{\hat{x}}$ corresponds to choosing $\hat{x}$ as an estimate for X .

$$f_{X|Y}(x|y) dx = P[H_x | Y=y]$$

↳ MAP estimator chooses an estimate $\hat{X}_{\text{MAP}}(y) = x$ to maximize the prob. of H_x given the observation $Y=y$

$$f_{X|Y}(x|y) = \frac{f_{Y|X}(y|x) f_X(x)}{f_Y(y)} = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

↳ Since denominator $f_Y(y)$ doesn't depend on x , maximizing $f_{X|Y}(x|y)$ is equivalent to maximizing $f_{Y|X}(y|x) f_X(x)$

If we don't have access to apriori probabilities $f_X(x)$, we ignore that term and obtain the maximum likelihood (ML) estimator

Def 9.2: The ML estimate of X given observation Y is

$$\hat{X}_{ML}(y) = \operatorname{argmax}_x f_{Y|X}(y|x)$$

In terms of cont. of hypotheses H_x and the experimental observation $y < Y \leq y+dy$, we observe that

$$f_{Y|X}(y|x) dy = P[y < Y \leq y+dy | H_x]$$

★ ML is equivalent to MAP when all possible values of X are equally likely.

When X and Y are discrete RV with sample values in the sets S_X and S_Y given observation $Y = y_i$

$$\rightarrow \hat{X}_{MAP}(y_i) = \operatorname{argmax}_{x \in S_X} P_{Y|X}(y_i|x) P_X(x)$$

$$\rightarrow \hat{X}_{ML}(y_i) = \operatorname{argmax}_{x \in S_X} P_{Y|X}(y_i|x)$$

For discrete RVs, MAP maximizes the prob. of choosing the correct x_i , but, in general, one may perform better than the other depending on the application.

Example 21:

Consider an experiment that produces a Bernoulli random variable with probability of success q . In this experiment, q is a sample value of RV Q with PDF

$$f_Q(q) = \begin{cases} 6q(1-q) & , 0 \leq q \leq 1 \\ 0 & , \text{o.w} \end{cases} \rightarrow \text{Beta}(2, 2) \text{ BV}$$

$$\hookrightarrow \text{Beta}(i, j) = \frac{1}{B(i, j)} x^{i-1} (1-x)^{j-1}$$
$$\frac{B(i, j)}{\Gamma(i)\Gamma(j)} = \int_0^1 x^{i-1} (1-x)^{j-1} dx$$

To estimate q , we perform n ind. Bernoulli experiments. The number of successes in n trials is K . Given the observation $K=k$, derive Q estimators

a) Blind estimate $\hat{q}_B$

$$\hat{q}_B = E[Q] = \int_0^1 q f_Q(q) dq$$

$$\text{Mean of } \beta \text{ function: } E[\text{Beta}(i, j; x)] = \int_0^1 x \frac{x^{i-1} (1-x)^{j-1}}{B(i, j)} dx = \frac{B(i+1, j)}{B(i, j)}$$

$$\hat{q}_B = \frac{2}{2+2} = \frac{1}{2} \quad , \quad = \frac{\frac{\Gamma(i+1)\Gamma(j)}{\Gamma(i+j+1)}}{\frac{\Gamma(i)\Gamma(j)}{\Gamma(i+j)}} = \frac{i}{i+j}$$

~~$$E[\text{Beta}^2(i, j; x)] = \frac{B(i+2, j)}{B(i, j)} = \frac{i(i-1)}{(i+j+1)(i+j)}$$~~

~~$$\text{Var}[\text{Beta}(i, j; x)] = \frac{i}{i+j} \left(\frac{i-1}{i+j+1} - \frac{i}{i+j} \right) = \frac{i(i^2 - i^2 + ij - i^2 - ij - i)}{(i+i+1)(i+i+2)} =$$~~

b) Maximum Likelihood Estimate $\hat{q}_{ML}(k)$

To find other estimates, we observe in the problem that for any $Q=q$, K is a binomial RV. Therefore, cond. PMF of K is

$$P_{K|Q}(k|q) = \binom{n}{k} q^k (1-q)^{n-k}$$

The ML estimate is the value of q that maximizes $P_{K|Q}(k|q)$

$$\frac{dP_{K|Q}(k|q)}{dq} = \binom{n}{k} q^{k-1} (1-q)^{n-k-1} [k(1-q) - (n-k)q] = 0$$

$$k - nq = 0$$

$$\hat{q}_{ML}(k) = \frac{k}{n}$$

c) MAP estimator $\hat{q}_{MAP}(k)$

MAP estimator is the value of q that maximizes

$$f_{Q|K}(q|k) = \frac{P_{K|Q}(k|q) f_Q(q)}{P_K(k)} \rightarrow \text{const wrt } q$$

$$\frac{d}{dq} [P_{K|Q}(k|q) f_Q(q)] = \frac{d}{dq} \left[6 \binom{n}{k} q^{k+1} (1-q)^{n-k+1} \right]$$

$$= 6 \binom{n}{k} q^k (1-q)^{n-k} [(k+1)(1-q) - (n-k+1)q] = 0$$

$$k - kq + 1 - q - nq + kq - q = 0$$

$$\hat{q}_{MAP} = \frac{k+1}{n+2} //$$

d) The minimum mean square error estimate $\hat{q}_M(k)$

$$\hat{q}_M(k) = E[Q | k=k] \rightarrow \frac{P_{K|Q}(k|q) f_Q(q)}{P_K(k)} \rightarrow \text{have to find this}$$

$$P_K(k) = \int_{-\infty}^{\infty} \underbrace{P_{K|Q}(k|q) f_Q(q)}_{f_{K,Q}(k,q)} dq$$

$$= \int_0^1 b \binom{n}{k} q^{k+1} (1-q)^{n-k+1} dq = b \binom{n}{k} \beta(k+2, n-k+2)$$

$$f_{Q|K}(q|k) = \begin{cases} \frac{q^{k+1} (1-q)^{n-k+1}}{\beta(k+2, n-k+2)} & , 0 \leq q \leq 1 \\ 0 & , \text{o.w.} \end{cases} \quad \left. \vphantom{\frac{q^{k+1} (1-q)^{n-k+1}}{\beta(k+2, n-k+2)}}} \right\} \text{Beta}(k+2, n-k+2)$$

$$\Rightarrow \hat{q}_M(k) = E[Q | k=k] = \frac{i}{i+j} = \frac{k+2}{n+4}$$

e) Optimum Linear Estimate $\hat{q}_L(k)$

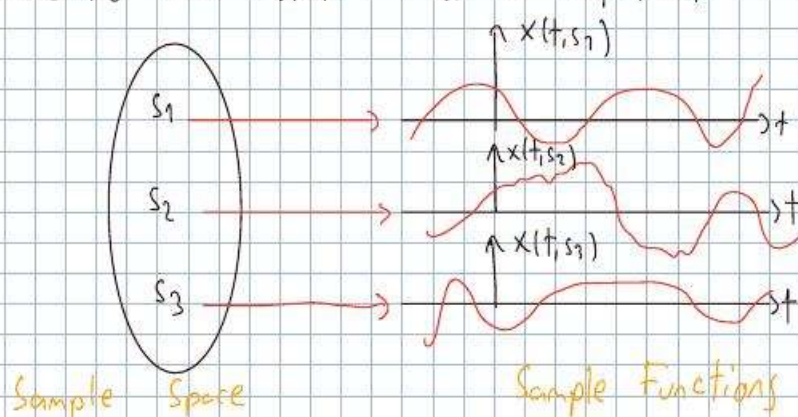
Note that MMSE estimate $\hat{q}_M(k) = \frac{1}{\underbrace{n+4}_{a^*}} k + \frac{2}{\underbrace{n+4}_{b^*}}$ is already a linear funct of k with

$$\text{Thus } \hat{q}_L(k) = \hat{q}_M(k)$$

CHAPTER 10: STOCHASTIC PROCESSES

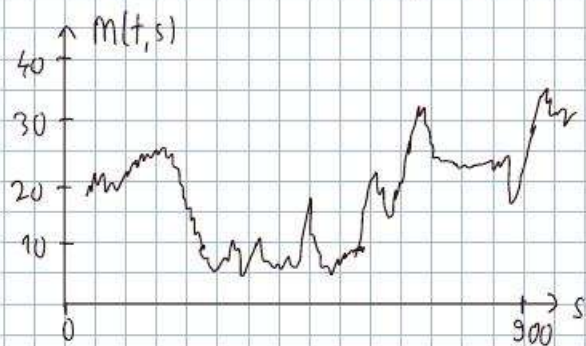
Stochastic means random, process means function of time. We'll study random functions of time.

Definition: A stochastic process $X(t)$ consist of an experiment with probabilities defined on a sample space S and a function that assigns a time function $x(t,s)$ to each outcome $s \in S$. A sample function $x(t,s)$ is the function associated with outcome s of an experiment



The ensemble of a stochastic process is the set of all possible time functions that can result from an experiment.

Example 22: Consider an experiment in which we record $M(t)$, the number of active calls at a telephone switch at time t , at each second over a time interval of 15 minutes



$\Rightarrow$ One possible observation of $M(t)$

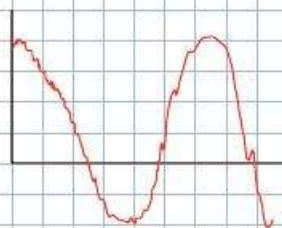
Types of Stochastic Processes

Definition: $X(t)$ is a discrete-value process if the set of all possible values $X(t)$ at all times is a countable set S_X ; otherwise it is a continuous value process.

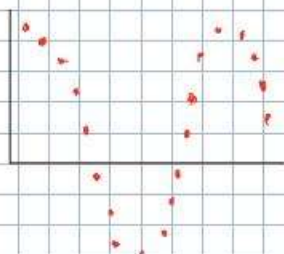
Definition: $X(t)$ is a discrete-time process if it is defined only for a set of time instants $t_n = nT$, where T is a constant and n is an integer; otherwise $X(t)$ is a continuous value process.

* Random sequence X_n is an ordered seq of variables: $X_0, X_1, \dots$

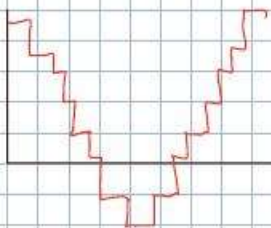
Cont. time, cont value



Disc. time, cont. value



Cont. time, disc value



Disc time, disc value

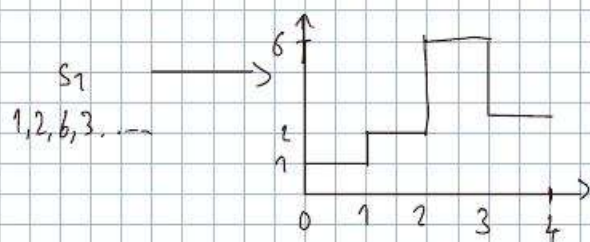


Random Variables From a Random Process

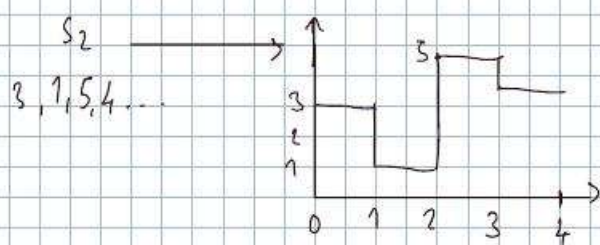
Suppose we observe a random process at time $t = t_1$. Each time we perform the experiments, we observe a sample function $x(t, s)$ and that sample function determines the value which is $x(t_1, s)$. Since s is random $x(t, s)$ is also random $\Rightarrow$ We can use the notation $X(t_1)$ to denote the corresponding random variable

If we sample a process $X(t)$ at k time instants $t_1, \dots, t_k$, we obtain the k -dimensional random vector $\mathbf{X} = [X(t_1), \dots, X(t_k)]'$

Example 23: Suppose that we roll a die at $T=0,1,2,3$ and record the output, N_T , where $1 \leq N_T \leq 6$. Define $X(t)$ as $X(t) = N_T$ for $T \leq t < T+1$.



Find PMF of $X(3.5)$
 ↳ The value of $X(3.5)$ is equal to the value of N_3 which has uniform PMF



$$P_{X(3.5)}(x) = \begin{cases} \frac{1}{6} & , x = 1, \dots, 6 \\ 0 & , \text{o.w} \end{cases}$$

Independent Identically Distributed Random Sequences

Theorem 28: Let X_n be an iid random sequence $X_n = X(nT)$. For a discrete-value process, the sample $\underline{X} = [X_1 \ X_2 \ \dots \ X_k]^T$ has joint PMF

$$P_{\underline{X}}(\underline{x}) = P_X(x_1) P_X(x_2) \dots P_X(x_k) = \prod_{i=1}^k P_X(x_i)$$

Similarly for a continuous-value process, the joint PDF is

$$f_{\underline{X}}(\underline{x}) = f_X(x_1) f_X(x_2) \dots f_X(x_k) = \prod_{i=1}^k f_X(x_i)$$

Definition: A Bernoulli(p) process X_n is an iid random sequence in which each X_i is a Bernoulli(p) random variable

Example 24:

For a Bernoulli(p) process X_n , find the joint PMF of $\underline{X} = [X_1 \dots X_n]'$

For a Bernoulli(p) RV, $P_X(x_i) = \begin{cases} p^{x_i} (1-p)^{1-x_i} & , x_i \in \{0,1\} \\ 0 & , \text{o.w} \end{cases}$ ★

$$P_X(\underline{x}) = \prod_{i=1}^n P_X(x_i) = \prod_{i=1}^n p^{x_i} (1-p)^{1-x_i} = p^k (1-p)^{n-k}$$

where $x_i \in \{0,1\}$ and $k = x_1 + x_2 + \dots + x_n$

$$P_X(\underline{x}) = \begin{cases} p^{x_1 + \dots + x_n} (1-p)^{n-x_1-x_2-\dots-x_n} & , x_i \in \{0,1\} \\ 0 & , \text{o.w} \end{cases}$$

Poisson Process

Definition: A stochastic process $N(t)$ is a counting process if for every sample function, $n(t,s) = 0$ for $t < 0$ and $n(t,s)$ is integer-valued and non-decreasing with time

If we count the number of customers arriving at interval $(t_0, t_1]$ the result is $N(t_1) - N(t_0)$

Definition: A counting process $N(t)$ is a Poisson process of rate λ if

- The # of arrivals at any interval $(t_0, t_1]$, $N(t_1) - N(t_0)$, is a Poisson random variable with expected value $\lambda(t_1 - t_0)$
- For any pair of nonoverlapping $(t_0, t_1]$ and $(t'_0, t'_1]$, the number of arrivals in each interval, $N(t_1) - N(t_0)$ and $N(t'_1) - N(t'_0)$ respectively are independent random variables.

Buraya Theorem 10.2, 10.3, 10.4 yazılacak

Expected Value and Correlation

Defn: The expected value of a random process $X(t)$ is

$$\mu_X(t) = E[X(t)]$$

Example 25: If R is a nonnegative RV, find the expected value of $X(t) = R |\cos 2\pi f t|$

$$\mu_X(t) = E[X(t)] = E[R \cdot |\cos 2\pi f t|] \rightarrow \text{Expected value over samples of } X(t) \text{ Not over time points of } x_i(t)$$

Definition: The autocovariance function of $X(t)$ is

$$C_X(t, \tau) = \text{Cov}[X(t), X(t+\tau)]$$

The autocovariance function of sequence X_n is

$$C_X[m, k] = \text{Cov}[X_m, X_{m+k}]$$

$$\star \begin{aligned} C_X(t, 0) &= \text{Cov}[X(t), X(t)] = \text{Var}[X(t)] \\ C_X[m, 0] &= \text{Cov}[X_m, X_m] = \text{Var}[X_m] \end{aligned}$$

Definition: The autocorrelation function of $X(t)$ is

$$R_X(t, \tau) = E[X(t)X(t+\tau)]$$

The autocorrelation function of X_n is

$$R_X[m, k] = E[X_m X_{m+k}]$$

Theorem 30:

$$C_X(t, \tau) = R_X(t, \tau) - \mu_X(t) \mu_X(t + \tau)$$

$$C_X[m, k] = R_X[m, k] - \mu_X[m] \mu_X[m + k]$$

Example 26: Given a random process $X(t)$ with expected value $\mu_X(t)$ and autocorrelation $R_X(t, \tau)$, define $Y(t) = X(t) + N(t)$, where $N(t)$ is a noise process with $\mu_N(t) = 0$ and autocorrelation $R_N(t, \tau)$.

Assuming that $N(t)$ is independent of $X(t)$, find the exp. value and autocorrelation of $Y(t)$.

$$\mu_Y(t) = E[Y(t)] = E[X(t)] + E[N(t)] = \mu_X(t)$$

$$\begin{aligned} R_Y(t, \tau) &= E[Y(t) Y(t + \tau)] = E[(X(t) + N(t))(X(t + \tau) + N(t + \tau))] \\ &= E[X(t) X(t + \tau)] + E[N(t) N(t + \tau)] + E[X(t) N(t + \tau)] + E[X(t + \tau) N(t)] \\ &= R_X(t, \tau) + R_N(t, \tau) + E[X(t)] E[N(t + \tau)] + E[X(t + \tau)] E[N(t)] \\ &= R_X(t, \tau) + R_N(t, \tau) \end{aligned}$$

Stationary Processes

At any time t_1 , we sample a random process $X(t)$ and obtain a random variable $X(t_1)$ with PDF $f_{X(t_1)}(x)$. In general, this PDF is a function of t_1 and in general

$$f_{X(t_1)}(x) \neq f_{X(t_2)}(x) \quad \text{for } \forall t_1, t_2 \text{ s.t. } t_1 \neq t_2$$

HOWEVER

for stationary processes $f_{X(t_1)}(x)$ does not depend on t_1 . That is for any two time instants t_1 and $t_1 + \tau$

$$f_{X(t_1)}(x) = f_{X(t_1 + \tau)}(x) = f_X(x)$$

Definition: A stochastic process $X(t)$ is **stationary** if and only if for all sets of time instants $t_1, \dots, t_m$ and any time difference τ

$$f_{X(t_1) \dots X(t_m)}(x_1, \dots, x_m) = f_{X(t_1 + \tau) \dots X(t_m + \tau)}(x_1, \dots, x_m)$$

A random sequence X_n is **stationary** if and only if for all sets of integer time instants $n_1, \dots, n_m$ and integer difference k ,

$$f_{X_{n_1} \dots X_{n_m}}(x_1, \dots, x_m) = f_{X_{n_1 + k} \dots X_{n_m + k}}(x_1, \dots, x_m)$$

Theorem 3.1: For a stationary process $X(t)$, we can show the following

a) $\mu_X(t) = \mu_X$

b) $R_X(t, \tau) = R_X(0, \tau) = R_X(\tau)$

c) $C_X(t, \tau) = R_X(\tau) - \mu_X^2 = C_X(\tau)$

Example 27: Consider a random process $X(t) = A \cos(2\pi f_c t + \theta)$

$$f_\theta(\phi) = \begin{cases} 1/2\pi, & 0 \leq \phi < 2\pi \\ 0, & \text{o.w} \end{cases}$$

↳ uniformly distributed phase

Find the exp value and autocorrelation of $X(t)$

$$\begin{aligned} \mu_X(t) &= E[X(t)] = E[A \cos(2\pi f_c t + \theta)] = A E[\cos(2\pi f_c t + \theta)] \\ &= A \int_0^{2\pi} \frac{1}{2\pi} \cos(2\pi f_c t + \phi) d\phi = \frac{A}{2\pi} \sin(2\pi f_c t + \phi) \Big|_0^{2\pi} \end{aligned}$$

$$\mu_X(t) = 0 = \mu_X$$

$$R_X(t, \tau) = E[X(t) X(t+\tau)] = A^2 E[\cos(2\pi f_c t + \theta) \cos(2\pi f_c (t+\tau) + \theta)]$$

$$= A^2 \frac{1}{2} \left[E[\cos(2\pi f_c \tau)] + E[\cos(2\pi f_c (2t+\tau) + 2\theta)] \right]$$

Zero since $\frac{1}{2} \sin[2\pi f_c (2t+\tau) + 2\theta] \Big|_0^{2\pi} = 0$

$$R_X(t, \tau) = \frac{A^2}{2} \cos(2\pi f_c \tau) = R_X(\tau)$$

↳ $X(t)$ has the properties of a stationary process but this does not necessarily mean that it is a stationary process

★ Narrow sense stationary:

↪ If μ_X and $R_X(\tau)$ time-indep, $X(t)$ is not necessarily stationary
only WSS

Definition: $X(t)$ is a wide-sense stationary (WSS) process iff
 $E[X(t)] = \mu_x$ and $R_x(t, \tau) = R_x(0, \tau) = R_x(\tau)$
time-indep time indep

X_n is a WSS sequence iff
 $E[X_n] = \mu_x$ and $R_x[n, k] = R_x[0, k] = R_x[k]$

All stationary processes satisfy this cond but if a process satisfying these conditions, we can't claim it's stationary

Example 28: For the process given in Ex. 27 we have $E[X(t)] = 0 = \mu_x$ and $R_x(t, \tau) = \frac{A^2}{2} \cos 2\pi f_c \tau = R_x(\tau)$
 $\hookrightarrow X(t)$ is a WSS process.

Theorem 32: For a WSS process $X(t)$

- a) $R_x(0) \geq 0$
- b) $R_x(\tau) = R_x(-\tau)$
- c) $R_x(0) \geq |R_x(\tau)|$

PF

a) $R_x(0) = E[X(t)X(t+0)] = E[X^2(t)] \geq 0$

b) $R_x(\tau) = E[X(t)X(t+\tau)] = E[X(t+\tau)X(t)]$
 $= R_x(t+\tau, -\tau) = R_x(-\tau)$

c) $E[(X(t) \pm X(t-\tau))^2] \geq 0$
 $E[X^2(t) \pm 2X(t)X(t-\tau) + X^2(t-\tau)] \geq 0$

$$\underbrace{E[X^2(t)]}_{R_x(0)} + \underbrace{E[X^2(t-\tau)]}_{R_x(0)} \geq \pm 2 \underbrace{E[X(t)X(t-\tau)]}_{R_x(\tau)}$$

$$R_x(0) \geq \pm R_x(\tau) \Rightarrow R_x(0) \geq |R_x(\tau)|$$

Example 29: Which of the following functions are valid autocorrelation functions? (for WSS)

	R_1	R_2	R_3
$R_1(\tau) = e^{- \tau }$	✓	✓	✓
$R_2(\tau) = e^{-\tau^2}$	✓	✓	✓
$R_3(\tau) = e^{-\tau} \cos \tau$	✓	✗	✗ $\rightarrow R_3(-2\pi) > R_3(0)$
$R_4(\tau) = e^{-\tau^2} \sin \tau$	✓	✗	✗ $\rightarrow R_4(\frac{\pi}{2}) > R_4(0)$

Definition The cross-correlation of cont. time random processes $X(t)$ and $Y(t)$ is

$$R_{X,Y}(t, \tau) = E[X(t)Y(t+\tau)]$$

The cross-correlation of random sequences X_n and Y_n is

$$R_{X,Y}[m, k] = E[X_m Y_{m+k}]$$

Definition: $X(t)$ and $Y(t)$ are jointly WSS if $X(t)$ and $Y(t)$ are individually WSS and $R_{X,Y}(t, \tau) = R_{X,Y}(\tau)$

Random seq. X_n and Y_n are jointly WSS if X_n & Y_n are individually WSS and $R_{XY}[m, k] = R_{XY}[k]$

Example 30:

X_n is a WSS random sequence with $R_X[k]$. Y_n is obtained by reversing the sign of every other RV in Y_n , i.e., $Y_n = (-1)^n X_n$

a) Express $R_Y[n, k]$ in terms of $R_X[n, k]$

b) Express $R_{XY}[n, k]$ in terms of $R_X[n, k]$

c) Is Y_n WSS?

d) Are X_n and Y_n jointly WSS?

$$\begin{aligned} \text{a) } R_Y[n, k] &= E[Y_n Y_{n+k}] = E[(-1)^n X_n (-1)^{n+k} X_{n+k}] \\ &= (-1)^{2n+k} E[X_n X_{n+k}] = (-1)^k R_X(k) \end{aligned}$$

↓ don't be fooled by $2n$

$$\begin{aligned} \text{b) } R_{XY}[n, k] &= E[X_n Y_{n+k}] = E[X_n (-1)^{n+k} X_{n+k}] \\ &= (-1)^{n+k} R_X[k] \end{aligned}$$

→ Not jointly WSS since a function of n d)

$$\text{c) } R_Y[n, k] = (-1)^k R_X[k] = R_Y[k] \quad \checkmark \quad \text{Cond 2}$$

$$\mu_Y[n] = E[Y_n] = E[(-1)^n X_n] = (-1)^n \mu_X \rightarrow \begin{array}{l} \text{constant if } \mu_X = 0 \\ \text{otherwise it's not} \\ \text{const} \end{array} \quad \text{Cond 1}$$

$\Rightarrow Y_n$ is WSS if $\mu_X = 0$

Theorem 33:

$$\begin{aligned} R_{XY}(\tau) &= R_{YX}(-\tau) \\ R_{XY}[k] &= R_{YX}[-k] \end{aligned}$$

Pf.

$$\begin{aligned} R_{xy}(\tau) &= E[X(t)Y(t+\tau)] \\ &= E[Y(t+\tau)X(t)] = R_{yx}(t+\tau, -\tau) = R_{yx}(-\tau) \end{aligned}$$

Definition: $X(t)$ is a Gaussian stochastic process if $\underline{X} = [X(t_1) X(t_2) \dots X(t_k)]'$ is a Gaussian random vector for all $k > 0$ and time instants $t_1, t_2, \dots, t_k$.

Theorem 34: If $X(t)$ is a Gaussian WSS random process then $X(t)$ is Gaussian stationary process.

Definition: $W(t)$ is a white Gaussian noise process if $W(t)$ is a stationary Gaussian random process with

$$\mu_w = 0 \text{ and } R_w(\tau) = \delta(\tau)$$

→ Every time instant has ∞ infinite correlation with itself
→ 0 correlation with each other

CHAPTER 11: RANDOM SIGNAL PROCESSING

For deterministic inputs, the response of LTI system is well studied

input / deterministic signal: $v(t)$, LTI filter response: $h(t)$

$$\Rightarrow w(t) = \int_{-\infty}^{\infty} h(u) v(t-u) du = \int_{-\infty}^{\infty} h(t-u) v(u) du$$

→ If input $X(t)$ is random, then output is

$$Y(t) = \int_{-\infty}^{\infty} h(u) X(t-u) du = \int_{-\infty}^{\infty} h(t-u) X(u) du$$

Theorem 35: $E[Y(t)] = E\left[\int_{-\infty}^{\infty} h(u) X(t-u) du\right] = \int_{-\infty}^{\infty} h(u) E[X(t-u)] du$

Theorem 36: If the input to LTI filter $h(t)$ is WSS, the output $Y(t)$ has the following properties

a) $Y(t)$ is WSS with

$$\mu_Y = E[Y(t)] = \mu_X \int_{-\infty}^{\infty} h(u) du$$

$$R_Y(\tau) = \int_{-\infty}^{\infty} h(u) \int_{-\infty}^{\infty} h(v) R_X(\tau+u-v) dv du$$

Proof

$$R_Y(t, \tau) = E[Y(t) Y(t+\tau)]$$

$$= E\left[\int_{-\infty}^{\infty} h(u) X(t-u) du \int_{-\infty}^{\infty} h(v) X(t+\tau-v) dv\right]$$

$$= E\left[\int_{-\infty}^{\infty} h(u) \int_{-\infty}^{\infty} h(v) X(t-u) X(t+\tau-v) du dv\right]$$

$$= \int_{-\infty}^{\infty} h(u) \int_{-\infty}^{\infty} h(v) E[X(t-u) X(t+\tau-v)] du dv$$

$$= \int_{-\infty}^{\infty} h(u) \int_{-\infty}^{\infty} h(v) R_X(\tau - v + u) du dv = R_Y(\tau) \Rightarrow Y(t) \text{ also WSS}$$

b) $X(t)$ & $Y(t)$ are jointly WSS with

$$R_{XY}(\tau) = \int_{-\infty}^{\infty} h(u) R_X(\tau - u) du = E \left[X(t) \int_{-\infty}^{\infty} h(u) X(t+\tau-u) du \right]$$

c) $R_Y(\tau) = \int_{-\infty}^{\infty} h(-w) R_{XY}(\tau - w) dw$

pf $R_Y(\tau) = \int_{-\infty}^{\infty} h(u) \underbrace{\int_{-\infty}^{\infty} h(v) R_X(\tau - v + u) dv}_{R_{XY}(\tau + u)} du = \int_{-\infty}^{\infty} h(u) R_{XY}(\tau + u) du$
 $\xrightarrow{w = -u} \int_{-\infty}^{\infty} h(-w) R_{XY}(\tau - w) dw$

Example 37: $X(t)$ a WSS process with exp. value $\mu_X = 10$ Volts is input to

$$h(t) = \begin{cases} e^{t/0.2} & , 0 \leq t \leq 0.1 \text{ s} \\ 0 & , \text{o.w} \end{cases}$$

What is the exp. value of the output process $Y(t)$

$$\mu_Y = \mu_X \int_{-\infty}^{\infty} h(u) du = 10 \int_0^{0.1} e^{5u} du = 2(e^{0.5} - 1) = 1.30 \text{ V}$$

Theorem 37: If $X(t)$ is a static Gaussian random process, then output $Y(t)$ is also a static Gaussian random process

Example 32: Let $h(t)$ be a low-pass filter with impulse response.

$$h(t) = \begin{cases} e^{-t} & , t \geq 0 \\ 0 & , \text{o.w} \end{cases}$$

The input $X(t)$ is WSS with $\mu_X = 2$ and $R_X(\tau) = \delta(\tau)$. What are the expected value and auto correlation of $Y(t)$?

$$\mu_Y = \mu_X \int_{-\infty}^{\infty} h(u) du = 2 \int_{-\infty}^{\infty} e^{-u} du = 2$$

$$R_Y(\tau) = \int_{-\infty}^{\infty} h(u) \int_{-\infty}^{\infty} h(v) \delta(\tau - v + u) dv du = \int_{-\infty}^{\infty} h(u) h(\tau + u) du$$

$$\text{For } \tau > 0, R_Y(\tau) = \int_0^{\infty} e^{-u} e^{-\tau - u} du = e^{-\tau} \int_0^{\infty} e^{-2u} du = \frac{e^{-\tau}}{2}$$

$$\text{For } \tau < 0, R_Y(\tau) = \int_{-\tau}^{\infty} e^{-u} e^{-\tau - u} du = e^{-\tau} \int_{-\tau}^{\infty} e^{-2u} du = \frac{e^{-\tau}}{2}$$

$$R_Y(\tau) = \frac{e^{-|\tau|}}{2}$$



Power-spectral Density of a Random Process

The power spectral density (PSD) function, $S_X(f)$, provides frequency domain representation of the time structure of $X(t)$

Definition:

$$S_X(f) = \lim_{T \rightarrow \infty} \frac{1}{2T} E[|X_T(f)|^2]$$

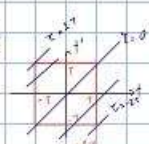
$$= \lim_{T \rightarrow \infty} \frac{1}{2T} E \left[\left| \int_{-T}^T X(t) e^{-j2\pi ft} dt \right|^2 \right]$$

Theorem 38 : Wiener - Khintchine

If $X(t)$ is a WSS process,

$$S_x(f) = \mathcal{F}\{R_x(\tau)\} = \int_{-\infty}^{\infty} R_x(\tau) e^{-j2\pi f\tau} d\tau$$

$$R_x(\tau) = \mathcal{F}^{-1}\{S_x(f)\} = \int_{-\infty}^{\infty} S_x(f) e^{+j2\pi f\tau} df$$



$$E[|X_T(t)|^2] = E\left[\int_{-\infty}^{\infty} X(t) e^{j2\pi ft} dt \int_{-\infty}^{\infty} X^*(t') e^{-j2\pi ft'} dt'\right] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} E[X(t)X^*(t')] e^{j2\pi f(t-t')} dt dt'$$

$$= \int_{-\infty}^{\infty} R_X(t-t') e^{j2\pi f(t-t')} dt' = 2T \int_{-\infty}^{\infty} R_X(\tau) \left|e^{-j\frac{\tau}{2T}}\right| e^{j2\pi f\tau} d\tau$$

$$\rightarrow S_X(f) = \lim_{T \rightarrow \infty} \frac{1}{2T} E[|X_T(f)|^2] = \int_{-\infty}^{\infty} R_X(\tau) e^{-j2\pi f\tau} d\tau = S_X(f)$$

Theorem 39: $S_X(f)$ has the following properties

- $S_X(f) \geq 0, \forall f$
- $\int_{-\infty}^{\infty} S_X(f) df = E[X^2(t)] = R_X(0) : \text{average power}$
- $S_X(-f) = S_X(f)$

Proof

- a) $S_X(t)$ is the expected value average of a nonnegative quantity

$$b) R_x(z) = \int_{-\infty}^{\infty} S_x(f) e^{j2\pi f z} df$$

$$R_x(0) = E^2[X(t)] = \int_{-\infty}^{\infty} S_x(f) df$$

$$c) S_x(f) = \int_{-\infty}^{\infty} R_x(\tau) e^{-j2\pi f\tau} d\tau = \int_{-\infty}^{\infty} R_x(\tau) e^{-j2\pi(-f)(-\tau)} d\tau$$

Define $\tau' = -\tau \Rightarrow S_x(f) = \int_{-\infty}^{\infty} \underbrace{R_x(-\tau')}_{R_x(\tau')} e^{-j2\pi(f)\tau'} d\tau' = S_x(-f)$

Example 33: PSD of $X(t)$ is $S_X(f)$

$$S_X(f) = \frac{5}{w} \operatorname{rect}\left(\frac{f}{2w}\right) = \begin{cases} 5/w, & -w \leq f \leq w \\ 0, & \text{o.w.} \end{cases}$$

a) What is the average power of $X(t)$?

$$R_X(0) = \int_{-\infty}^{\infty} S_X(f) df = \int_{-w}^w \frac{5}{w} df = 10 \text{ Watt}$$

b) Find $R_X(\tau)$

$$S_X(f) = 10 \frac{1}{2w} \operatorname{rect}\left(\frac{f}{2w}\right) \Rightarrow R_X(\tau) = \mathcal{F}^{-1}\{S_X(f)\} = 10 \operatorname{sinc}(2w\tau) \\ = 10 \frac{\sin(2\pi w\tau)}{2\pi w\tau}$$

Definition: For jointly WSS processes $X(t)$ and $Y(t)$, the cross spectral density is

$$S_{XY}(f) = \int_{-\infty}^{\infty} R_{XY}(\tau) e^{-j2\pi f\tau} d\tau$$

Example: Let $Y(t) = X(t) + N(t)$, where $N(t)$ is WSS process with $\mu_N = 0$. When $X(t)$ and $N(t)$ are jointly WSS, find $S_Y(f)$

$$R_Y(t, \tau) = E[Y(t)Y(t+\tau)] = E[(X(t)+N(t))(X(t+\tau)+N(t+\tau))] \\ = E[X(t)X(t+\tau)] + E[N(t)N(t+\tau)] + E[X(t)N(t+\tau)] + E[N(t)X(t+\tau)] \\ = R_X(\tau) + R_N(\tau) + R_{XN}(\tau) + R_{NX}(\tau) = R_Y(\tau) \Rightarrow \text{indep. at } t$$

$$E[Y(t)] = E[X(t)] + E[N(t)] = \mu_X = \mu_Y \Rightarrow \text{Then } Y(t) \text{ is also WSS}$$

$$S_Y(f) = \int_{-\infty}^{\infty} R_Y(\tau) e^{-j2\pi f\tau} d\tau = S_X(f) + S_N(f) + S_{NX}(f) + S_{XN}(f)$$

Frequency Domain Filter Relationship

If $v(t)$ is a deterministic signal,

Theorem 40: When the WSS process $X(t)$ is input to LTI filter with TF $H(f)$, the spectral density function of the output process $Y(t)$ is

$$S_Y(f) = |H(f)|^2 S_X(f)$$

Pf:

$$R_Y(\tau) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(u)h(v) R_X(\tau - u - v) du dv$$

$$S_Y(f) = \mathcal{F}\{R_Y(\tau)\} = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(u)h(v) R_X(\tau - u - v) du dv \right] e^{-j2\pi f\tau} d\tau$$

Define $\tau' = \tau - u - v$

$$\begin{aligned} S_Y(f) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(u)h(v) R_X(\tau') e^{-j2\pi f\tau'} e^{j2\pi fu} e^{j2\pi fv} du dv d\tau' \\ &= \underbrace{\int_{-\infty}^{\infty} h(u) e^{j2\pi fu} du}_{H^*(f)} \underbrace{\int_{-\infty}^{\infty} h(v) e^{j2\pi fv} dv}_{H(f)} \underbrace{\int_{-\infty}^{\infty} R_X(\tau') e^{-j2\pi f\tau'} d\tau'}_{S_X(f)} \\ &= |H(f)|^2 S_X(f) \end{aligned}$$

Example 35: A WSS process $X(t)$ with $R_X(\tau) = e^{-4\pi\tau^2}$ is the input to an LTI filter with TF

$$H(f) = \begin{cases} 1 & , \quad 0 \leq |f| \leq 2 \\ 0 & , \quad \text{o.w.} \end{cases}$$

Find

a) Average power of $X(t)$

$$E[X^2(t)] = R_X(0) = 1$$

b) $S_Y(f) = ?$

$$S_Y(f) = |H(f)|^2 S_X(f) \quad , \quad S_X(f) = \mathcal{F}\{R_X(\tau)\} = \frac{1}{2} e^{-\pi f^2/4}$$

$$S_Y(f) = \begin{cases} \frac{1}{2} e^{-\pi f^2/4} & , \quad |f| \leq 2 \\ 0 & , \quad \text{o.w.} \end{cases}$$

$$c) E[Y^2(t)] = R_Y(0) = \int_{-\infty}^{\infty} S_Y(f) df = \frac{1}{2} \int_{-2}^2 e^{-\pi f^2/4} df$$

Linear Estimation of Continuous Statistical Processes

We observe a sample function of WSS random process $Y(t)$ and design a linear filter to estimate a sample function of another WSS RP $X(t)$. The linear filter that minimizes the MSE is referred as the **Wiener filter**.

Theorem 4.2: $X(t)$ and $Y(t)$ are WSS RPs with PSDs $S_X(f)$ and $S_Y(f)$ respectively. They have cross-spectral density function $S_{XY}(f)$. $X(t)$ is the output of a linear filter with input $Y(t)$ and TF $H(f)$. The TF that minimizes the MSE $E[(X(t) - \hat{X}(t))^2]$ is

$$H(f) = \begin{cases} \frac{S_{XY}(f)}{S_Y(f)} & , \quad S_Y(f) \neq 0 \\ 0 & , \quad \text{o.w.} \end{cases}$$

$$e_L = \int_{-\infty}^{\infty} \left[S_X(f) - \frac{|S_{XY}(f)|^2}{S_Y(f)} \right] df$$

→ One classical problem is to separate a signal from additive noise.

$$\text{Let } Y(t) = X(t) + N(t)$$

where $X(t)$ and $N(t)$ are independent.

$$S_Y(f) = S_X(f) + S_N(f) \quad \text{and} \quad S_{XY}(f) = S_X(f)$$

$$\Rightarrow H(f) = \frac{S_X(f)}{S_X(f) + S_N(f)}$$

$$e_L = \int_{-\infty}^{\infty} \left(S_X(f) - \frac{|S_X(f)|^2}{S_X(f) + S_N(f)} \right) df$$

$$= \int_{-\infty}^{\infty} \frac{S_X(f) S_N(f)}{S_X(f) + S_N(f)} df$$

Example 36: $X(t)$ WSS RP with $\mu_X = 0$ and $R_X(\tau) = \frac{\sin(2\pi 5000\tau)}{2\pi 5000\tau}$

Observe $Y(t) = X(t) + N(t)$ with $N(t)$ WSS RP with $S_N(f) = 10^{-5}$
 $X(t)$ and $N(t)$ are mutually indep. Find the transfer function of the linear filter that minimizes MSE.

$$S_X(f) = \mathcal{F}\{R_X(\tau)\} = \begin{cases} 10^{-4}, & |f| \leq 5000 \\ 0, & \text{o.w} \end{cases}$$

$$H(f) = \frac{10^{-4}}{10^{-4} + 10^{-5}} = \frac{1}{1.1}, \quad |f| \leq 5000 \Rightarrow \text{LP filter}$$

Chapter 12: MARKOV CHAINS

Definition: A discrete-time Markov chain $\{X_n \mid n=0,1,\dots\}$ is a discrete time random sequence s.t. given $X_0, \dots, X_n$; the next RV depends only on X_n through

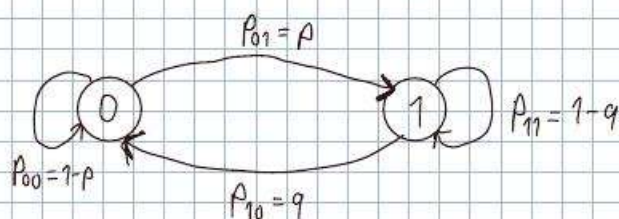
$$P[X_{n+1}=j \mid X_n=i_n, X_{n-1}=i_{n-1}, \dots, X_0=i_0] = P[X_{n+1}=j \mid X_n=i_n] = P_{ij}$$

X_n summarizes all past history of a system needed to predict the next variable X_{n+1}

X_n : State of the system and the sample space of RV X_n is called the state space

Theorem 43: $P_{ij} \geq 0$, $\sum_{j=0}^{\infty} P_{ij} = 1$

Example 37: Two-state Markov Chain



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